The Conversion of Cultural Tastes into Social Network Ties[1]

Kevin Lewis
University of California, San Diego

Jason Kaufman
McLean Hospital

In recent years, sociologists have focused less on cultural tastes as "effects" of social structure and more on their causal efficacy in the creation and maintenance of social ties. Progress on this agenda has been hindered, however, by limitations in theory, methods, and available data. This article attempts to advance all three fronts. First, it clarifies, integrates, and expands upon prior work to develop a more comprehensive theoretical framework for examining the conversion of cultural tastes into social relationships. Second, it introduces a powerful network modeling tool, stochastic actor-based modeling, that is uniquely capable of implementing this framework. Third, it illustrates the utility of these advances using an original, longitudinal data set based on the behavior of a cohort of college students on Facebook. Findings from this application suggest several general substantive propositions about capital conversion, providing a starting point for future research.

INTRODUCTION

Tastes play a fundamental role in the reproduction of class structure (Bourdieu 1984). This insight has launched a thriving field of research investigating both the origins of tastes in terms of social structural position (e.g., Peterson and Kern 1996; Mark 1998; Katz-Gerro 1999; Alderson, Junisbai, and

[1] We are indebted to Marco Gonzalez, Chana Teeger, Jocelyn Viterna, and Andreas Wimmer for providing valuable feedback on an earlier draft of this paper. We also thank Ron Breiger and Kurt Gray for helpful conversations regarding tie strength and network evolution, Marco Gonzalez for developing our exogenous measures of tastes, and Omar

0002-9602/2018/12306-0004$10.00

Heacock 2007; Bennett et al. 2009) and the consequences of tastes for a variety of socioeconomic outcomes (e.g., DiMaggio 1982; DiMaggio and Mohr 1985; Hagan 1991; Erickson 1996; Kaufman and Gabler 2004). One particularly innovative branch of this research takes a social network approach to the study of tastes (Pachucki and Breiger 2010, pp. 214–15). This work has demonstrated that the relationship between cultural tastes and social inequality is mediated by the impact of tastes on social network ties—in other words, by a process of capital conversion between cultural and social capital (DiMaggio and Mohr 1985; Bourdieu 1986; Lizardo 2006; Edelmann and Vaisey 2014).

While pioneering, we argue that this network-based approach to tastes and class reproduction has yet to realize its full potential. This is primarily due to four obstacles. First, prior research on tastes and networks tends to focus on tastes in a single domain (most commonly, music) and measure them in a particular way (most commonly, using genres). Closed-ended instruments facilitate data collection and comparison but risk imposing a structure on respondents' tastes (Marsden and Swingle 1994): respondents may prefer very different items within the same genre and may not conceptualize their tastes as genres in the first place (Lewis et al. 2008*b*). And while there are practical and theoretical reasons to focus on music, the possibility that the conversion value of preferences may vary by domain is seldom explored—leading to a potential overgeneralization of findings.

Second, prior research commonly relies on classifications of tastes into categories such as "highbrow" and "popular" that refer to the content of a preference and are held to be universally valid. Meanwhile, there is evidence that other dimensions of valuation—particularly those wedded to the local context—are equally important (Erickson 1996; Eliasoph and Lichterman 2003). We propose that one dimension of a taste's meaning that may be particularly salient for tie formation is its location in the local cultural ecology (Mark 1998, 2003; Kaufman 2004). For instance, if classical music lovers tend to befriend one another, is this because a taste for Bach is a universal marker of high status and exclusion (Lamont and Lareau 1988)? Or because it is relatively rare in the immediate social context and for this reason effectively distinguishes "us" from "them" (DiMaggio 1987; Lizardo 2006)? Because surveys often rely on random, nationally representative samples, portraits of local cultural ecologies are challenging to construct. Consequently,

Lizardo and Steve Vaisey for their communication and encouragement. An earlier version of this paper was presented at the 2015 Annual Meeting of the American Sociological Association in Chicago. This research was supported by National Science Foundation grant SES-0819400, Social Networks and Online Spaces: A Cohort Study of American College Students. Direct correspondence to Kevin Lewis, Department of Sociology, University of California, San Diego, 9500 Gilman Drive, 0533, La Jolla, California, 92093-0533. E-mail: lewis@ucsd.edu

some effects of tastes on ties may so far have been masked, while other findings may have been artifacts of omitted dimensions of meaning.

Third, past research tends to focus on only one mechanism whereby cultural capital is converted into social capital: the more tastes two individuals share, the more likely they are to form a tie. However, certain tastes may affect the tendency to form ties in general (cf. Goodreau, Kitts, and Morris 2009; Wimmer and Lewis 2010), and tastes that are not shared may be just as important (Edelmann and Vaisey 2014), such that two people with similar overall "taste profiles" are especially likely to become friends (Selfhout et al. 2009; see also Dahlander and McFarland 2013). Failure to control for alternative mechanisms could result in misdiagnosis of the conversion processes at work and underappreciation of the diverse ways tastes actually influence ties. However, it is impossible to disentangle these mechanisms unless one collects "sociocentric" network data on some closed population of respondents (Steglich, Snijders, and Pearson 2010), whereas many network data sets contain only "egocentric" data reported by a sample of disconnected individuals (e.g., Marsden 1987; Vaisey and Lizardo 2010).

Finally, most analyses of tastes and ties are cross-sectional—creating an uphill battle for causal claims given that tastes both influence and are influenced by social networks. While researchers have employed sophisticated techniques (e.g., Lizardo 2006) to address this concern, an important possibility has yet to be raised: What if conversion dynamics change over time at different stages in the evolution of a network? If they do change, this places additional qualifications on prior results and presents a further opportunity for theoretical development.

In light of the above limitations, this article develops a broader theoretical framework for examining the conversion of cultural tastes into social network ties. Building upon a growing body of work focusing on the structure rather than the content of cultural systems (Mohr 1998), this framework considers both "exogenous" dimensions of meaning dependent on universal classifications and "endogenous" dimensions of meaning derived from the local cultural ecology. We also offer theoretical grounds for expecting that tastes in different domains will be differentially consequential; we elaborate three general mechanisms whereby tastes might influence ties; and, drawing on classic and contemporary work in the networks literature (Verbrugge 1977; de Klepper et al. 2010), we suggest that different domains, meanings, and mechanisms may be more or less salient over time.

To implement this framework, we draw on recent developments in methods and data. First, assessing the impact of tastes on ties requires a modeling framework capable of (1) realistically representing and disentangling various possible tie-generating mechanisms; (2) handling the complex interdependencies of social network data (as opposed to traditional approaches that assume observations are independent); and (3) accomplishing these goals longitudinally (to minimize concerns of reciprocal causation). Stochastic actor-based models—a tool network analysts have developed over the past two decades but cultural sociologists have only begun to adopt (Lewis, Gonzalez, and Kaufman 2012; Edelmann and Vaisey 2014; see also Steglich, Snijders, and West 2006)—meet these requirements (Steglich et al. 2010).

Second, we illustrate the utility of this framework and method using a unique, longitudinal data set based on the activity of a cohort of college students on Facebook. Not only does this data set provide sociocentric network data on a naturally bounded population, but it also contains open-ended data on students' tastes that can be coded according to any number of (exogenous and endogenous) metrics. We have demonstrated in a previous article that friendship among these students is significantly associated with taste similarity (Lewis et al. 2008*b*)—an example of the ubiquitous social phenomenon known as network autocorrelation, homogeneity, or homophily (Steglich et al. 2010; Wimmer and Lewis 2010). Here we pursue an increasingly common agenda of delving beneath this observed pattern to understand how and under what circumstances it is generated and maintained (Rivera, Soderstrom, and Uzzi 2010).

This article is structured as follows: We begin by briefly positioning our work vis-à-vis prior research on capital conversion. Next, we develop our theoretical framework and introduce our method, data, and cultural coding schemes. We then present two sets of statistical models that document the impact of tastes on network evolution—first by considering all data collectively and next by differentiating effects across time periods. We find that, controlling for exogenous dimensions of meaning, tastes are also efficacious if they are particularly common or rare among one's peers; that tastes have both positive and negative effects on the likelihood of forming and maintaining relationships; and that "dyadic conversion," "generalized conversion," and "cultural matching" are all mechanisms whereby tastes are converted into ties. However, different conversion mechanisms, taste domains, and dimensions of meaning are consequential at different stages in the college experience as originally "invisible" preferences gradually emerge. We extrapolate from these results several general substantive propositions about capital conversion. While the generalizability of these propositions remains to be explored—a problem familiar to network analysts (Borgatti et al. 2009)—and our data face limitations of nonresponse and interpretation—problems endemic to online research (Golder and Macy 2014)—we hope our theoretical approach, method, and findings will provide a starting point for future inquiry.

THE CAPITAL CONVERSION MODEL

Traditionally, the dominant conceptualization of culture and networks has been to treat "cultural practices and patterns of culture consumption and taste as being primarily *shaped* and *determined* by social networks" (Lizardo 2006, p. 779; emphasis in original). For some time, however, the pieces were in place for an alternative approach. Lamont and Lareau advocated greater attention to "micro-level interactions in which individuals activate their cultural capital to . . . attain desired social results" (1988, p. 163). Bryson pointed out that we "do not really know how people use taste in their everyday lives" (1996, p. 897). These comments align with a broader theoretical agenda emphasizing culture's causal role in the evolution of networks (Emirbayer and Goodwin 1994; DiMaggio 2004; Pachucki and Breiger 2010) and in the determination of action more generally (Alexander and Smith 2001; Jacobs and Spillman 2005; Vaisey 2009).

While some evidence existed for the role of tastes in tie formation (Erickson 1996; Ostrower 1998; Long 2003), these strands had not yet been integrated into an explicit causal framework. Drawing on Bourdieu's classic essay (1986), Lizardo (2006) argued that tastes can be utilized to form and maintain social ties through a process of "capital conversion." Specifically, because popular culture is shared by wider swaths of the population and "provides the stuff of everyday sociability" (DiMaggio 1987, p. 444), it fosters the creation of weak tie "bridges" across social space. Meanwhile, because high culture is more strongly associated with social position, it can be used to identify insiders from outsiders and build strong tie "fences" among the elite.

This work paved an exciting new path at the intersection of cultural sociology and network analysis—one that complements prior work on stratification and demonstrates that "culture" is not only transmitted through network ties but also helps create and sustain them (see also Selfhout et al. 2009; Vaisey and Lizardo 2010; Lewis et al. 2012; Edelmann and Vaisey 2014). Unfortunately, this work also inherited the challenges of its intellectual forebears. Culture, in all its many forms, is notoriously difficult to operationalize (Mohr and Ghaziani 2014), and it is still relatively recently that network analysts have developed both the language (Rivera et al. 2010) and the tools (Snijders 2011) for understanding the complex processes whereby social networks evolve (Lewis 2015*a*). In the next section, we clarify and expand on existing models of capital conversion in four ways—related to media, meaning, mechanisms, and time—to propose a broader theoretical framework for considering how tastes impact ties.

A NEW THEORETICAL FRAMEWORK

Media

There is a common tendency in past work on tastes—both related and unrelated to capital conversion—either to focus on a single domain of preference or else to utilize aggregate measures that consider multiple domains simultaneously (for examples of the latter, see Hughes and Peterson [1983], Katz-Gerro [1999], Mark [2003], Lizardo [2006], and Alderson et al. [2007]; examples of the former are ubiquitous). And yet there are a number of reasons to expect that whether or how tastes influence relationships may critically vary by medium.[2]

Tastes in music, for instance, are highly correlated with socioeconomic status (although the nature of this relationship has famously changed over time; Peterson 1992) and provide nuanced markers of subcultural distinction (Roy and Dowd 2010). As such, they are a natural site for examining culture (e.g., Marsden and Swingle 1994)—but also one that may be unrepresentative of other domains. In an unusually thorough study of multiple taste domains in Britain, Bennett et al. (2009) found that tastes in movies correlate less strongly with class but are sharply divided by age and gender. Meanwhile, reading books is a relatively unpopular activity uniquely associated with formal education (cf. Bukodi 2007). We might expect, then, that these three domains imply very different criteria for deciding whether to befriend a stranger: one might care *what* someone listens to; *whether* someone reads; and only whether someone likes movies *appropriate* to her age and gender. While the connotations associated with each domain are obviously contextually and historically contingent (e.g., Baumann 2001; Griswold, McDonnell, and Wright 2005; Zavisca 2005), the implications for capital conversion are clear: the medium of a taste fundamentally shapes its social signal and thus its potential impact on tie formation.

The above distinctions have to do with how tastes "directly" influence interaction, that is, by making some people more or less attractive as potential friends. But tastes may also impact ties indirectly by serving as "foci" around which joint activities are organized (Feld 1981; Benediktsson 2012). For instance, roommates might gather to watch their favorite movies, strangers could meet at a concert for their favorite band, and book clubs provide opportunities for periodic interaction among friends (see Long 2003). Colleges are rife with cultural organizations, and websites such as Meetup explicitly connect strangers with common interests. All of these scenarios entail self-selection into a situation in which additional time is spent with one's (culturally similar) companions or additional opportunities are present to meet (like-minded) strangers. Ceteris paribus, we can expect the importance of a domain for capital conversion to vary with the availability of domain-specific, institutionalized foci that regularly organize interaction.[3]

[2] While in principle this discussion could be extended to any domain—preferred wine, preferred travel destinations, preferred pets—we focus here on media, both to foreshadow our available measures (on movies, music, and books) and to emphasize these tastes' particular importance to interaction and identity (e.g., Bennett et al. 2009).

[3] The importance of such foci cannot be overstated. For instance, Kossinets and Watts (2009) demonstrate that even a modest preference to affiliate with similar others, compounded over many "generations" of self-selection into similar social foci and proximate network positions, can result in striking patterns of community segregation.

Meaning

Work on cultural classification has typically relied on what we call "exogenous" measures of preferences. In other words, tastes are coded according to their content using categories such as "highbrow" and "popular" that are held to be universally valid. Such an approach is employed even in studies that move beyond this basic dichotomy: the traditional definition of "omnivore," for instance, is one whose tastes span the "elite" versus "nonelite" boundary (Peterson 1992; Warde and Gayo-Cal [2009] call this "omnivorousness by composition"). To complement this strategy of measuring meaning, we propose an alternative grounded in two additional perspectives. First, scholars in the tradition of microsociology have long emphasized locally constituted meaning and action (e.g., Eliasoph and Lichterman 2003; Fine and Fields 2008). Because meaning is grounded in specific social settings, the utility of cultural resources depends on the context in which they are used (Erickson 1996). But what features of a context are especially consequential?

Here we draw upon a second perspective emphasizing the location of each taste in the broader cultural system of which it is part (see Mohr 1998). This "ecosystem" places limits on cultural stability and change, and it is these dynamics—rather than the substance of cultural forms per se—that are central to research on "cultural ecologies" (Kaufman 2004). Mark (1998, 2003), for instance, argues that musical forms compete for the time, energy, and preferences of individuals and in this way carve out niches in sociodemographic space. Lieberson (2000), meanwhile, demonstrates that tastes continuously evolve due to the desire for social differentiation. In other words, "some consumers desire novelty, though nothing so novel as to be unrecognizable vis à vis current trends. . . . A given taste's value thus lies in its relationship to the systemwide distribution of tastes" (Kaufman 2004, pp. 347–48).

Research on cultural ecologies is therefore premised on the notion that people are sensitive to the relative novelty or commonality of available preferences. However, rather than using this sensitivity to explain the tastes we choose to identify with, we consider its consequences for the people we choose to affiliate with. The above research therefore serves as the background (cultural) context in which the focal (social) dynamics of this article play out: we expect tastes to affect ties not only because they are universal signals of a particular type of culture, but also because tastes carry distinct social meanings derived from how frequently they occur in the local setting. We refer to this popularity as the endogenous meaning of a preference, and our proposed theoretical framework recognizes that both exogenous and endogenous properties of tastes may be responsible for network evolution.[4]

[4] Notably, although operationalized in terms of high and popular culture, the essence of Lizardo's argument—and the DiMaggio piece that inspired it—is also ecological in na-

Mechanisms

Facilitated by advances in statistical modeling, recent research on social networks has focused less on patterns in relationships and more on the underlying mechanisms that generated them (Rivera et al. 2010; Lewis 2015*a*). A fundamental contribution of this work has been to caution against assuming a direct correspondence between process and pattern. For instance, simply because two people with the same racial background are friends does not necessarily mean that the reason they are friends is because they have the same background (Goodreau et al. 2009; Wimmer and Lewis 2010). Here we adapt this contribution to the study of tastes, both by incorporating insights from network research as well as by clarifying (and more directly measuring) mechanisms cultural sociologists have already posited.

Dyadic Conversion

Prior work has focused almost exclusively on one micromechanism whereby cultural tastes are converted into social relationships: the more tastes two individuals share, the more likely it is they will form a tie. While sometimes qualifications regarding the nature of the shared taste or the strength of the social relationship are specified (e.g., DiMaggio 1987; Lizardo 2006), this general mechanism has been developed in a variety of empirical applications. In short, commonalities in preferences, knowledge, experiences, and worldviews provide the shared social "glue" that fosters social interaction and sustains a variety of long-term relationships (DiMaggio and Mohr 1985; Kalmijn 1994; Erickson 1996; Vaisey and Lizardo 2010; Lewis et al. 2012). As noted above, dyadic conversion can also operate indirectly by enlarging opportunities for joint activities (Werner and Parmelee 1979; Benediktsson 2012): the more tastes two people share, the more likely they are to turn up in the same place at the same time, dramatically increasing their likelihood of meeting (Feld 1981, 1982).

---

ture. Lizardo acknowledges that "from an ecological viewpoint . . . highbrow culture is simply an example of a *specialist* form, and its effect on relational outcomes should be due to this latter property and not to its higher status value" (2006, p. 801, emphasis in original). DiMaggio also observes that "conversations about *scarce* cultural goods bind partners who can reciprocate and identify as outsiders those who do not command the required codes" (1987, p. 443; emphasis added). We build on these contributions by developing the distinction between exogenous and endogenous meanings and suggesting that each has consequences for networks *net* of the other. Unfortunately, because we do not have data on tie strength (or data on interactions from which to infer it; e.g., Arnaboldi, Guazzini, and Passarella 2013), we are unable to pursue Lizardo's agenda of distinguishing the consequences of tastes for different kinds of relationships. We revisit this issue in the conclusion.

Generalized Conversion/Inhibition

We also expect that individuals with certain tastes may accumulate more or fewer relationships overall (cf. Goodreau et al. 2009; Wimmer and Lewis 2010). In other words, tastes should not be considered a restricted type of capital that only supports ties with others who share it. Rather, certain tastes might constitute a form of generalized capital (cf. Bearman 1997) that fosters tie formation with anyone, while other tastes may generally stifle network growth. This mechanism has received the least attention in the literature, although it squares most closely with the notion that tastes—an accumulated cultural resource—may be converted into another kind of broad resource: the quantity of one's social connections (Campbell, Marsden, and Hurlbert 1986; see also DiMaggio and Mohr 1985, p. 1256). Specifically, if there is indeed a status hierarchy of preferences—whether entire domains or specific tastes and whether universally or locally ordered—the distribution of friendships should be patterned accordingly: people with esteemed tastes should be generally more popular, while people with stigmatized tastes may face social marginalization as a result (see Steglich et al. 2006, p. 50). We refer to these mechanisms as "generalized conversion" and "generalized inhibition," respectively.[5]

Cultural Matching

Homophily—the principle that "birds of a feather flock together"—has been studied across a wide range of characteristics and relationships (McPherson, Smith-Lovin, and Cook 2001). While work on cultural homophily is less common, this concept has featured centrally in simulation-based studies of group dynamics (e.g., Carley 1991; Axelrod 1997) and is increasingly the focus of empirical research (e.g., Rivera 2012). While dyadic conversion is also consistent with the concept of homophily, we recognize homophily can be operationalized in many ways (see DiMaggio and Garip 2012, p. 111) and see a need to distinguish between the effects of sharing specific cultural attributes and resembling someone holistically. Following Rivera (2012), we refer to the latter as "cultural matching."

How is matching distinct from other mechanisms? Unlike generalized conversion—an effect that makes certain individuals more or less likely to form ties—both matching and dyadic conversion have to do with the likelihood of a tie forming within a given dyad. While dyadic conversion focuses

[5] It is worth noting that insofar as individuals with shared tastes select into particularly "greedy" foci—social groups that demand exclusive loyalty and time (Coser 1974)—we should observe the distinct signature of *social closure* from a capital conversion perspective: dyadic conversion coupled with generalized inhibition, i.e., tastes that bring people together at the cost of associating with anyone else.

on the quantity of shared tastes, however, matching focuses on the similarity of cultural profiles (cf. Selfhout et al. 2009; Dahlander and McFarland 2013). This distinction can be clarified using a hypothetical example. Suppose students A and B both like books by Kurt Vonnegut—so much so that they refuse to read anything else. Student C, meanwhile, loves not only Vonnegut but also Milan Kundera and Nicole Krauss. The dyadic conversion mechanism would predict that a tie between any of these students would be equally likely to develop: all three share a taste for Vonnegut, and this shared taste would help sustain conversation about the relative merits of *Slaughterhouse-Five* versus *Cat's Cradle*, joint participation in Vonnegut book clubs, and so on—all of which contribute to friendship. Cultural matching, meanwhile, would predict that a tie between A and B is more likely to develop than a tie between A and C or B and C, because the entirety of A and B's literary identities are defined by their passion for this one author. While dyadic conversion, therefore, predicts a linear association between the quantity of shared tastes and the likelihood of tie formation, matching predicts that a friendship between two people is more likely the more objectively similar their overall cultural profiles are.[6]

Time

While longitudinal analyses of networks have become more common in recent years (Snijders and Doreian 2010, 2012), surprisingly few studies have examined how the determinants of tie formation vary over time. To our knowledge, no prior work has examined such variation in capital conversion. We speculate, however, that different domains and dimensions of tastes may be more or less salient at different stages in the evolution of a network. Verbrugge (1977) is often credited for developing the distinction between "meeting" and "mating," respectively, the processes whereby strangers are converted to acquaintances and acquaintances to friends.[7] Recent work has further suggested that similarity with respect to "visible" characteristics (e.g., gender, age) is more important for meeting, while "invisible" characteristics (e.g., tastes) are more relevant for mating—because two peo-

[6] More formally, matching could also be interpreted in terms of balance theory (Cartwright and Harary 1956): if two students share both likes and dislikes, this entails balance on a greater number of dimensions and makes the formation of a friendship more likely than dyadic conversion alone would predict. On the other hand—and although Edelmann and Vaisey (2014) provide evidence that shared nonconsumption of musical genres is as important to tie formation as shared consumption—it is important to acknowledge that not liking a cultural object is different from actually disliking it, and our data cannot differentiate between the two.

[7] Verbrugge, in turn, points to several theoretical predecessors, including Lazarsfeld and Merton and social psychologists Levinger and Snoek (see Verbrugge 1977, n. 1).

ple need to be acquainted before they can determine each other's preferences (van Duijn et al. 2003; Selfhout et al. 2009; de Klepper et al. 2010).

Building on this observation, we propose that in relatively small and/or "constrained" social settings—that is, "where everyone is forced to interact much and often" (Feld 1981, p. 1019)—everyone will eventually "meet" sooner or later. Thus, the question is not whether but when different kinds of tastes will come into play. For instance, preferences that are particularly unusual, connected to institutionalized foci, or associated with outward lifestyle markers such as speech or dress might be relevant for relationship formation from the start. Meanwhile, whether through direct exposure (e.g., a chance meeting) or secondhand information (e.g., via friends-of-friends), in a small enough setting even originally "invisible" preferences should become visible to everyone with time, potentially triggering dormant conversion mechanisms.

Summary

Prior work on tastes and networks has tended to focus on tastes in a single domain (and so variation in the effects of tastes across domains has been obscured); on "exogenous" cultural measures (even when the underlying argument is ecological; e.g., Lizardo 2006); on a single mechanism (dyadic conversion) whereby tastes are converted into ties; and on processes that are undifferentiated with respect to time. The consequences of this are fourfold. First, insofar as there is a mismatch between sophisticated theoretical arguments and indirect empirical tests, support for these arguments is necessarily tentative.[8] Second, what appears to be a relationship between taste property X and network outcome Y could actually be an artifact of some other variable Z with which X is correlated—whether Z is an omitted dimension of meaning or an ignored domain of preference. Third, our limited understanding of capital conversion means that the scope conditions of this process have yet to be explored—leading to potential overgeneralization of conclusions to taste domains, social settings, and time horizons where they do not in fact hold. Fourth, prior work has potentially underappreciated the diverse ways tastes actually influence ties—impoverishing our understanding of culture and networks.

[8] For instance, due to constraints in available data, both Lizardo (2006) and Vaisey and Lizardo (2010) examine individual-level signatures of what are in fact dyadic mechanisms: the theoretical expectation that "certain kinds of people will form ties with one another" becomes the empirical prediction that "certain kinds of people will have certain types of networks." While the latter certainly follows from the former, so it also follows from alternative mechanisms; recent work has cautioned against precisely this kind of inference (e.g., Goodreau et al. 2009).

Just as limitations in available data and methods have stymied theoretical progress, so our own theoretical framework comes with a heavy price tag. Data sets containing both cultural and network variables are scarce to begin with. In order to implement our framework, relational data should also be collected for a relatively bounded setting, tastes should be documented in as unconstrained a format as possible, and both network and cultural data must be available longitudinally. Before explaining how our data set approaches these requirements, we first introduce an analytic method that can accommodate it.

STOCHASTIC ACTOR-BASED MODELS

In recent years, tremendous advances have been made in statistical models for network dynamics (Snijders 2011). Here we rely on the stochastic actor-based models described by Snijders (2001, 2005). In contrast to techniques for cross-sectional and/or egocentric data (fig. 1), these models allow us to estimate the processes responsible for the evolution of an entire social network over time—viewing global transformations in network structure as the accumulation of microlevel tie changes between actors. Such a framework is necessary to adequately disentangle the various mechanisms described above. In short, in order to say anything meaningful about the reasons two people became friends, it is necessary to compare these choices to the people who did not become friends (Steglich et al. 2010). A longitudinal approach also enables us to look at the impact of tastes on succeeding network changes, as opposed to documenting patterns in static network structures where the direction of causality is ambiguous (cf. Lewis et al. 2012).

Despite their great promise, applications of stochastic actor-based models among cultural sociologists remain surprisingly scarce (e.g., Lewis et al. 2012; Edelmann and Vaisey 2014). Because most readers will be unfamiliar with this method, we here summarize its defining characteristics (for an accessible introduction, see Snijders, van de Bunt, and Steglich 2010).

Overview and Assumptions

Previous models of network dynamics often focused on specific sets of micromechanisms and lacked explicit estimation theories, making them inadequate for multivariate theory testing (Snijders et al. 2010). Stochastic actor-based models, in contrast, can represent a wide variety of influences on network change and estimate parameters quantifying these influences. These models are characterized by four assumptions.

First, stochastic actor-based models conceive of network evolution as occurring in continuous time. In other words, rather than viewing each wave of data as a discrete "event" to be directly explained by the previous obser-

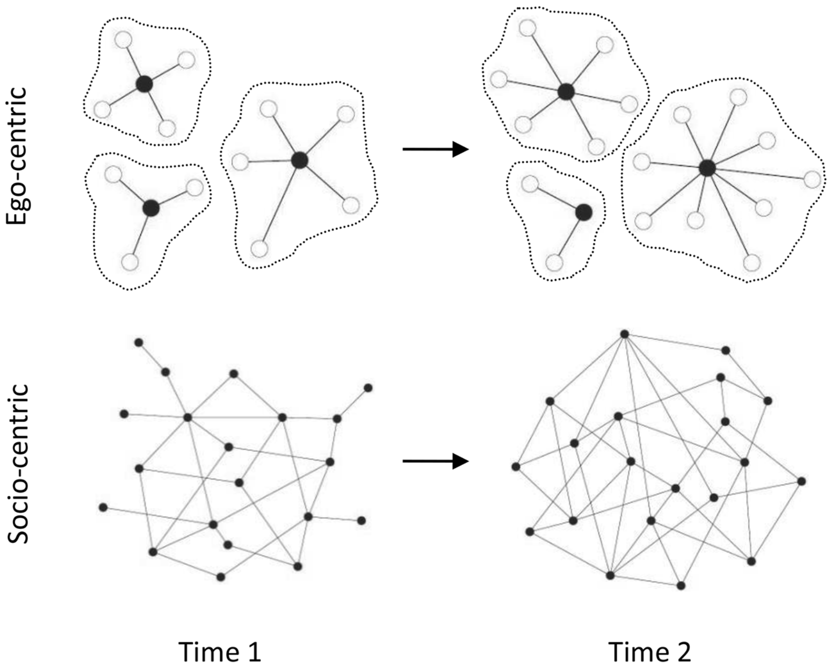


FIG. 1.—Different forms of longitudinal network data. With egocentric data—appropriate, for example, for random sampling—respondents (shaded nodes) are asked about their first-degree connections (e.g., friends), but it is unclear how or whether these friends are connected to one another or which individuals were not selected as friends. Sociocentric data, meanwhile, are typically collected from relatively small settings and include both ties and nonties as well as direct and indirect connections. Sociocentric data are necessary to effectively examine capital conversion but require sophisticated modeling techniques that relax the assumption of independence.

vation, these models simulate the process whereby the network at time K gradually evolves into the network at time $K + 1$. Each wave of data is therefore considered merely a "snapshot" of an underlying process of ongoing social change (see elaborations in Snijders et al. [2010], Steglich et al. [2010]).

Second, network changes are considered to be the outcome of a Markov process. In other words, the current state of the network, and only the current state of the network, probabilistically determines its further evolution. While likely unrealistic, this assumption has been made by practically all past models of network dynamics. Snijders et al. suggest considering it as a "lens" through which to view the data: "it should help but it also may distort" (2010, p. 46).

Third, actors control their outgoing ties. In other words, the decision to establish (or terminate) a tie is made exclusively by the actor who sends it on the basis of the specific considerations in the objective function (see below). This assumption of "structural individualism" is the reason for the name "actor-based model" because all changes in the network are driven by the behavior of individuals. (This is true even in our analysis of Facebook friendships, where models replicate the process whereby Facebook friendships actually develop: a tie is created if and only if a request is sent and then confirmed, while it may be dissolved by either actor.)

Fourth, network evolution is decomposed into its smallest possible components: "microsteps" in which a single social tie is modified (Steglich et al. 2006). Another way of putting this is that at any given "moment" one probabilistically selected actor is given the opportunity to create a new tie, dissolve an old one, or do nothing. No more than one tie can change at any one moment, and therefore tie changes may depend on each other only sequentially. This is another assumption that simplifies modeling considerably and is relatively benign in practice.

The Objective Function and Estimation

While the probability of receiving the opportunity to make a tie change can depend on attributes or network position (according to the rate function), we here assume it is equal for all actors for each transition period. Therefore, the sole function that needs to be specified is the "objective function," which has the following general shape:

$$f_i(\beta, x) = \sum_k \beta_k s_{ki}(x).$$

The objective function is the heart of the stochastic actor-based modeling approach. It determines the behavioral "rules" that actors will probabilistically follow when given the opportunity to make a change, or the short-term "objectives" they tend to pursue (whether purposefully or not). In the equation above, $f_i(\beta, x)$ is the value of the objective function for actor i depending on the state x of the network. The functions $s_{ki}(x)$ are effects, and the weights β_k are statistical parameters. Each effect represents a certain aspect of the network from the perspective of actor i. Some of these are derived from the characteristics of individual students, such as the tendency for women to form more or fewer friendships than men. Others are derived from characteristics of pairs of students, such as the tendency for students to befriend others who live in the same residence or share the same tastes.[9] Still others

[9] In order to calculate how various student traits influence the objective function, models must assume all students are aware of the traits of all other students in the sample—including their cultural tastes. On one hand, the ability of all students to see on Facebook the tastes of all other students makes this assumption more plausible than in other settings. On the other hand, this assumption will bias results if the distribution of tastes is

have nothing to do with student attributes but refer to "structural" network processes such as triadic closure (the tendency for A to befriend C if A is already friends with B and B is friends with C) and preferential attachment (the tendency for people with many friends to accumulate new friendships at a faster rate than people with fewer friends). Because multiple underlying processes (e.g., matching and triadic closure) can produce the same network pattern (e.g., the clustering of students by taste), it is essential to control for both in order to assess the actual contribution of each to observed behavior (Goodreau et al. 2009; Steglich et al. 2010; Wimmer and Lewis 2010).

Because of the complex dependencies between ties implied by the above processes, parameter values cannot be estimated directly. Instead, we estimate them using an approach called "method of moments" that depends on computer simulation (Snijders 1996, 2001). Unlike other modeling approaches that utilize cross-sectional data (e.g., Robins et al. 2007), the analysis does not make inferences about the determinants of network structure at the initial observation. Rather, simulations condition on the first wave of data, and it is the subsequent transition periods between waves that are modeled. If we recall that stochastic actor-based models conceive of network evolution in continuous time, then any given set of parameter values implies a certain (probabilistic) trajectory of development. The aim of model estimation is therefore to select, through a process of iterative refinement, coefficients where this simulated trajectory fits the observed data as closely as possible. In other words, it identifies the "movie" (parameter values) that best matches our array of "still frames" (panel waves).

THE "TASTES, TIES, AND TIME" DATA SET

Contemporary social media provide an unprecedented opportunity to collect unusually nuanced, "naturally occurring" cultural and network data at minimal cost (Golder and Macy 2014). To illustrate the utility of our theoretical framework, applied via stochastic actor-based models, we created a data set based on the activity of a cohort of American college students ($N =$ 1,640 at wave 1) on the popular social network website Facebook (Lewis et al. 2008b). With permission from Facebook and the college in question, from March 2006 (the students' freshman year) through March 2009 (the students' senior year) we downloaded profile and network data once per year and supplemented it with data provided by the college. Below we provide

inhomogeneous across the social foci where students most often meet. To assess this concern, we compared the means of all individual taste covariates (see table 1) by residence and major at each of the first three waves using one-way ANOVA tests. Out of 42 such tests (7 covariates × 2 foci × 3 waves), there were only two instances of significantly different means by residence and eight instances of significantly different means by major.

background on the study population and describe the individual and relational data used in this article.

Study Sample and Setting

The college used in this study is highly selective, and the vast majority of students live on campus all four years. While many students come from advantaged backgrounds, the college observes a need-blind admissions policy and makes special efforts to recruit a student body that is racially, socioeconomically, and geographically diverse (see Lewis et al. 2008b). Importantly, students at this college were also early adopters of Facebook: as of March 2006, 97.4% of students in our study cohort maintained an account. Here we focus on students who were members of the class of 2009 all four years (i.e., they did not transfer in or out of the college) and who had "publicly available" taste data for the first three years.[10] In this way, students whose profiles were set to "private" were excluded, as were students whose profiles were "public" but did not list any tastes.[11] This resulted in a final study sample of 520 students (31.7% of the original cohort).

Comparing these 520 students with the 1,120 students who were excluded, students in our analyses are more likely to be male ($P < .01$) and Hispanic ($P < .05$) and less likely to be Asian ($P < .01$). Otherwise, the two samples are statistically indistinguishable with respect to racial background, socioeconomic status, and online activity (as defined below). On the other hand, although membership on Facebook was nearly ubiquitous among this cohort, participation on Facebook—including listing one's tastes—is likely correlated with personality traits such as extraversion or general openness. This is particularly impactful because students without taste data were dropped. Combined with the fact that Facebook itself makes tastes more explicitly visible to one's peers than in face-to-face settings—and culture's impact on networks may be especially pronounced during the "unsettled" time that is college (Swidler 1986)—we should interpret this setting as a relatively liberal test of our capital conversion framework.[12]

[10] Because tastes are used here exclusively as "explanatory variables"—i.e., tastes at wave K are used to explain the network transition from wave K to wave $K+1$—taste data at wave 4 are superfluous.

[11] It is essential to remember that Facebook looked and functioned very differently in 2006 than it does today. One important difference is that available privacy settings have grown dramatically more complex. In 2006, one could essentially only make one's entire profile visible to (1) everyone, (2) only one's "network" (i.e., college), or (3) only one's friends. For the purposes of this article, the final category is considered "private" and these students were excluded. For an analysis of privacy behavior in this network, see Lewis, Kaufman, and Christakis (2008*a*).

[12] That said, other factors bias against identifying tastes' effects on networks using Facebook (see, e.g., n. 15), and all models include controls for proxy measures of online activity (see below).

Focusing on a subset of one cohort also has implications for the accuracy of results given that ties to people outside the sample are truncated. Given that no setting is perfectly isolated, this is a problem all network research faces to some degree, particularly when data are collected on a large scale. The most important way these missing data could be consequential for the current study is if observed ties that were facilitated by unobserved connections to third parties are mistakenly attributed to tastes, that is, triadic closure masquerading as homophily (Goodreau et al. 2009). Fortunately, our detailed data on social foci help protect against this possibility: because *all* interaction in college—both observed and unobserved in our data—is powerfully structured by the formal and informal settings in which students meet, much of the unobserved impact of third parties will likely be absorbed by our multiple measures of foci, described below.[13]

Demographic Background, Online Activity, and Shared Foci

In assessing our capital conversion framework, it is necessary to control for a number of alternative determinants of network dynamics (see review in Rivera et al. 2010). First, countless studies have documented the importance of demographic attributes for the evolution of networks. In particular, individuals tend to self-segregate on the basis of race, gender, and a variety of other characteristics (McPherson et al. 2001). Students from certain backgrounds may also form more or fewer ties overall or display varying degrees of "sociality" (Goodreau et al. 2009; Wimmer and Lewis 2010). Gender was coded based on self-report or first name and profile photo otherwise. Students were assigned one of five racial categories based on Facebook photos, surnames, and affiliation with Facebook groups signaling a racial identity. As a proxy for socioeconomic status, we combined self-reported hometown zip codes with socioeconomic data from the 2000 census and assigned each student the median household income of her zip code tabulation area. Finally, we coded each student's region of origin based on standard census divisions. Descriptive statistics of these variables are provided in table 1.

Second, if friendships on Facebook are to be used to approximate some underlying structure of relationships, we must control for the fact that some students are more active users of the site and may have a greater tendency to connect online. While all our models feature an endogenous control for this

[13] To further assess the possible impact of missing relational data on results, we replicated our most comprehensive model (model 2) without the control for triadic closure. Excluding this control (for observed instances of triadic closure) had generally small and inconsistent effects on the measurement of our central taste mechanisms—suggesting results would not have been substantially different if complete data on all instances of triadic closure had been available. We acknowledge, however, that this is an important limitation of our data set.

TABLE 1

MEAN (SD) FOR SELECT INDIVIDUAL AND DYADIC COVARIATES

	Wave 1	Wave 2	Wave 3
Individual covariates:			
Female	.45		
	(.50)		
Black[a]	.09		
	(.28)		
Asian[a]	.17		
	(.38)		
Mixed[a]	.03		
	(.18)		
Hispanic[a]	.08		
	(.26)		
Median household income (*K*)	65.47		
	(28.26)		
Days since last update	14.85		
	(29.31)		
Days since joined Facebook	253.79		
	(40.77)		
Consecrated culture	4.11	3.69	3.21
	(4.38)	(4.11)	(4.06)
Mass culture	7.76	6.45	5.77
	(6.99)	(6.14)	(5.75)
Specialized culture	8.29	7.75	6.97
	(9.82)	(8.66)	(7.88)
Common culture	7.81	5.50	4.00
	(6.03)	(4.75)	(3.77)
Movies	9.57	8.24	7.61
	(8.21)	(7.30)	(7.42)
Music	14.41	12.00	9.85
	(15.98)	(13.47)	(11.89)
Books	6.08	5.75	5.12
	(4.66)	(4.77)	(4.77)
Dyadic covariates:			
Shared major	.05		
	(.22)		
Shared Facebook group	.85		
	(1.29)		
Shared consecrated culture	.18	.15	.11
	(.50)	(.45)	(.38)
Shared mass culture	.45	.31	.24
	(.91)	(.73)	(.62)
Shared specialized culture	.01	.01	.01
	(.11)	(.09)	(.08)
Shared common culture	.72	.44	.29
	(1.22)	(.89)	(.68)
Shared movies	.29	.20	.16
	(.67)	(.54)	(.48)
Shared music	.46	.29	.18
	(1.13)	(.82)	(.62)
Shared books	.25	.19	.15
	(.55)	(.50)	(.44)
Taste similarity (%)	1.48	1.11	.85
	(2.16)	(1.96)	(1.72)

NOTE.—All of the above covariates have complete data ($N = 520$) except racial background indicators ($N = 518$) and median household income ($N = 466$). For dyadic covariates, the number of observations is equal to 269,880, i.e., $N^2 - N$ (a square matrix with the diagonal omitted).

[a] Dummy-coded variables, with "white" as reference category.

tendency ("preferential attachment"), we include two additional proxies for online activity in our analyses. Facebook has discontinued these features, but they were available on all student profiles at wave 1. The first indicated the date on which the user became a member of Facebook. The second indicated the date on which the user had last updated her profile. Using this information, we derived two measures of online activity ("days since joined Facebook" and "days since last update") that might reasonably be expected to influence tie formation.

Third, scholars have long emphasized the fundamental role that opportunity structures, or shared foci, play in structuring social networks (Feld 1982). For example, one of the most important determinants of friendship among college students is propinquity, such as sharing a room or dorm building (Marmaros and Sacerdote 2006; Wimmer and Lewis 2010). Friendships are also heavily influenced by institutional foci such as classrooms and academic majors (Mayer and Puller 2008; Kossinets and Watts 2009). While students at this college choose their own roommates after their freshman year—and therefore such choices cannot be used as exogenous predictors of friendship—all students are randomly assigned to one of several large residence facilities in which they spend their sophomore through senior years. The college provided data on such assignments for virtually all students as well as the academic major students declared at the end of their freshman year. (Unfortunately, class enrollment data were unavailable.)

Extracurricular activities are a final, especially important set of foci to consider. Not only do student publications, sports teams, and many kinds of social organizations provide opportunities to forge and strengthen friendships (Adamic and Adar 2003; Benediktsson 2012), but students are especially likely to self-select into these activities on the basis of their tastes. In this way, foci may mediate the impact of tastes on ties, as discussed in our theoretical framework. Because we were unable to acquire student organization rosters, we use an available proxy: at the time of our data collection, students formed a variety of online groups on Facebook that often—but not always—corresponded to formal student organizations ranging from religious groups to dance troupes to ethnic clubs. Other groups represented physical locations (e.g., dorm buildings), and many were purely electronic, signaling a common interest (e.g., political cause) or collective identity (e.g., the largest group in our data was for students who attended a public high school). All groups a student joined were listed on her profile. The average student in our sample belonged to 25 groups in 2006, and all students in the sample belonged to at least one.[14]

[14] Unfortunately, group data are only available at wave 1. At the time of data collection, it had not occurred to us to consider Facebook groups as social foci so we did not continue to collect these data.

Group data are limited in several ways. Some members of an organization may not be members of its Facebook group; not all organizations created a group in the first place; and not all groups represent opportunities for face-to-face meeting (although even shared "virtual" affiliations can inspire offline connections). We thus interpret group comembership as representing a variety of personal similarities and meeting opportunities—similarities and opportunities that may or may not overlap with other terms in our models. For this reason, we pause in our presentation of results to discuss findings from a model where all controls for shared foci are omitted in order to discern their impact on the estimation of other effects.

Cultural Data

The dominant tendency in research on tastes is to rely on closed-ended questions, commonly about music, and commonly asking respondents to choose from among a list of predetermined genres. While there are practical reasons for this practice, it also entails drawbacks: respondents may be thinking of entirely different items when they select the same genre, respondents may not (and in fact, our data suggest they generally do not) conceptualize tastes as genres in the first place (cf. van Venrooij 2009), and respondents will be miscategorized if a survey does not include all relevant options (Peterson 2005, p. 268; see also Bryson 1996).

Every Facebook profile contained open-ended spaces for users to list their favorite movies, music, and books. Online cultural data are not without limitations. First, favorites posted on Facebook may be as much a performance as an expression of authentic taste. Because we are interested in the social consequences of tastes, however, what is important is not so much what students actually prefer but what they express as tastes in public spaces. Favorites on Facebook seem to meet this criterion. Second, these measures would seem to capture only the extremes of cultural preference, ignoring variation among items that students "like" but would not necessarily consider their "favorites" (Schultz and Breiger 2010). However, if tastes influence the creation and maintenance of social ties, we would again expect that it is precisely these publicized favorites that are most personally meaningful and socially consequential.

Third, one limitation of our data set vis-à-vis data gathered from questionnaires is that observed tastes may be "out of date" if students do not update their profiles regularly. Fortunately, students at this university were not only early adopters of Facebook but unusually active users: at wave 1, for instance, half the students in our sample had last updated their profile in the past five days, 75% in the past two weeks, and 95% in the past 70 days. Facebook profiles were also structured very differently in 2006–9 such that "Favorites" appeared more prominently and were likely a greater focus of attention than they are today. Thus, it seems online and offline behaviors were coupled unusually strongly in this community during this time.[15]

Data on students' tastes underwent an extensive cleaning and coding process. First, responses were cleaned by hand to correct misspellings, remove superfluous commentary, and ensure all instances of the same taste were worded identically (e.g., "LOTR" and "Lord of the Rings").[16] Descriptive statistics on the three taste fields that resulted are presented in table 2. Across all waves and transition periods, the field of music displays the most participation (the quantity of students who expressed tastes), the least stability (the degree of consistency over time), and by far the most differentiation (the quantity of unique items). Meanwhile, the fields of movies and books display similar levels of participation, but the field of movies is slightly more differentiated and less stable than the field of books.

Second, all 10,385 distinct tastes in our data set were assigned a dichotomous code on each of four dimensions: consecrated culture, mass culture, specialized culture, and common culture (fig. 2). This approach acknowledges that a single taste can have multiple meanings (Erickson 1996; Eliasoph and Lichterman 2003) and accommodates individuals whose taste profiles are composed of diverse, or even dissonant, cultural elements (Lahire 2008; Ollivier 2008).

Exogenous Coding

While any number of exogenous coding schemes is possible, we employ two categories that have been examined in prior literature and are easily operationalized using objective measures: consecrated culture and mass culture.[17] We sought criteria that focus on particular cultural items, are specific

[15] A related problem is that tastes at wave K are used to predict the evolution of networks from wave K to wave $K + 1$—while in reality, students' preferences may change at any time during this interval. In this sense, our test for the effects of tastes that may be out of date by the end of the time interval is a conservative one.

[16] Two additional coding decisions were particularly consequential. First, while students tended to list movie *titles* and music *artists*, they listed both titles *and* authors for "Favorite Books." To ensure commensurability, all books were recoded according to their author. Second, all taste items that were part of a series were recoded as the title of that series. Otherwise, we would face a counterintuitive situation where one student who listed *Star Wars* and another who listed all of the individual *Star Wars* movies by name would be seen as having different tastes.

[17] In prior versions of this manuscript, we instead used the terminology "high culture" and "popular culture." These labels were potentially misleading, however, insofar as our measures of the former focus on consecration projects—the very inclusion in which may undermine a taste's value as a status token among the elite (cf. Macdonald 1983)—and our measures of the latter focus on the success of a cultural item, whereas "popular culture" commonly refers to items produced and distributed by mainstream commercial companies, independent of actual popularity.

TABLE 2
DESCRIPTIVE STATISTICS OF TASTE FIELDS AND NETWORK STRUCTURE

	Wave 1	Period 1	Wave 2	Period 2	Wave 3	Period 3	Wave 4
Taste fields:							
Participation (no. respondents):							
Movies	1,086		777		669		424
Music	1,107		802		696		470
Books	1,076		771		662		430
Differentiation (no. unique taste listings):							
Movies	1,918		1,717		1,577		1,353
Music	3,439		2,731		2,613		2,073
Books	1,599		1,338		1,233		1,018
Stability (avg. Jaccard coefficient):[a]							
Movies		.61		.70		.80	
Music		.50		.59		.72	
Books		.62		.70		.83	
Network structure:							
Statics:							
Average degree	35.54		49.25		55.51		58.03
Total no. ties	9,134		12,462		13,557		8,744
Proportion missing data[b]	.01		.03		.06		.42
Dynamics:							
Ties created (0 to 1)		3,656		1,794		878	
Ties dissolved (1 to 0)		200		125		81	
Ties maintained (1 to 1)		8,691		11,607		7,519	

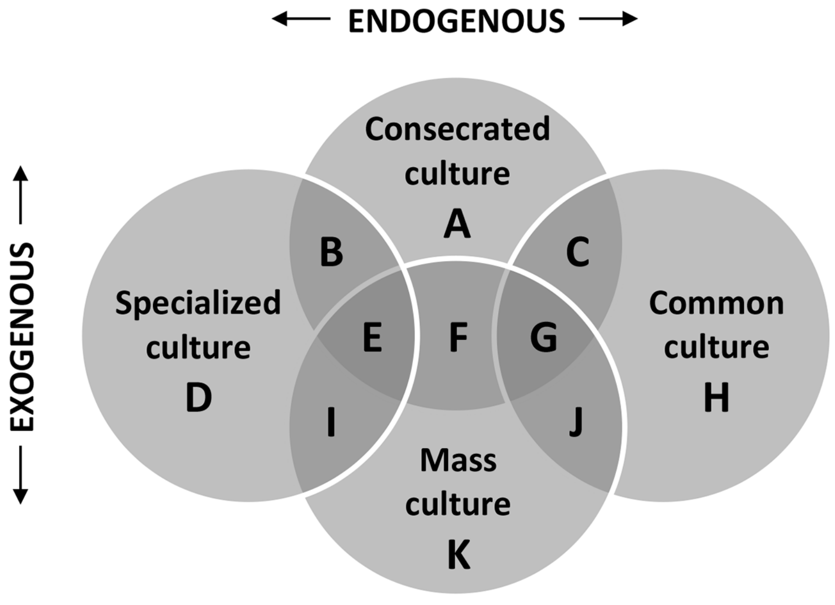
NOTE.—Descriptive statistics of taste fields are provided for all students in the original data set, under the assumption that this is the relevant reference point for local cultural meanings. Tastes at wave 4 are not employed in the current analyses (see n. 10). Descriptive statistics of network structure are provided only for students in the sample ($N = 520$).

[a] The Jaccard coefficient for "stability" is calculated identically to our dyadic measure of cultural matching—except applied to the same individual over time rather than two individuals from the same wave. In other words, it represents the quantity of tastes a student keeps constant over time divided by the total quantity of unique tastes the student expresses between both waves.

[b] While we only included in our analyses students with publicly available taste data (i.e., they did not have "private" profiles and they reported at least one taste) at all of the first three waves, students could separately set their friendship data to public or private—resulting in varying degrees of missing network data over time.

to the American context, and most closely correspond to the notion of consecrated culture as being critically legitimated as worthy of admiration and respect (Allen and Lincoln 2004) and mass culture as being widely consumed among the population at large (Peterson 1992).[18] All coding schemes

[18] In some sense, our definition of "mass culture" is endogenous insofar as it is based on the prevalence of a given cultural item. However, the frame of reference is the population at large rather than the immediate social context; and all of our mass culture measures emphasize the number of people who consume a given item rather than those who necessarily like it. It is also important to note that these definitions of consecrated and mass



	Favorite movies	Favorite music	Favorite books
A	Dr. Strangelove	Mozart	Ralph Ellison
B	Philadelphia Story	Cole Porter	Willa Cather
C	Shawshank Redemption	Beethoven	F. Scott Fitzgerald
D	Full Metal Jacket	Reel Big Fish	Howard Zinn
E	E.T.	Hank Williams	Sinclair Lewis
F	Schindler's List	Jimi Hendrix	Vladimir Nabokov
G	Lord of the Rings	Bob Dylan	J.D. Salinger
H	Fight Club	Coldplay	Jane Austen
I	Scary Movie	Monkees	Dean Koontz
J	Wedding Crashers	Beatles	J.K. Rowling
K	American Beauty	Eric Clapton	David Sedaris

FIG. 2.—Cultural classification scheme and examples of tastes from wave 1

were generated in 2008 to code data from waves 1 and 2 and updated in 2009 to code data from waves 3 and 4.[19]

First, a movie was considered consecrated culture if it has ever appeared on the American Film Institute's list of the top 100 films. A movie was con-

culture are not mutually exclusive. In fact, many items appeared on both lists across all three media, reflective of the fact that some consecrated culture becomes so canonized and popularized as to merit mass culture status (see fig. 2).

[19] Other operationalizations than those presented here are certainly possible. In many cases, the optimal measure was not available (e.g., to our knowledge, a list of the top-selling books of all time does not exist, nor does a list of popular movies based on ticket

sidered mass culture if it was among the highest grossing films of all time according to worldwide box office profits—the cutoff point being $200 million.

Second, National Public Radio maintains a list of the 100 most important American musical works of the 20th century. This list spans all genres of music. An artist was considered consecrated culture if she has ever been nominated for inclusion on this list or if she appeared among the 100 best classical artists according to a panel of experts at *Gramophone*, a magazine devoted to classical music. Using data from the Recording Industry Association of America, an artist was considered mass culture if she was listed among the top-selling artists in the United States in terms of album sales—the cutoff point being 10.25 million albums sold.[20]

Finally, an author was coded as consecrated culture if a book written by that author appeared on the Modern Library's list of the top 100 fiction and nonfiction books. This list was devised by the Modern Library's board members and is limited to works published in the English language during the 20th century. An author was considered mass culture if the author has ever had a book (fiction or nonfiction) reach the number one position on the *New York Times* best-seller list. This list dates back to August 9, 1942, when the first best-seller was named.

Endogenous Coding

In addition to the coding described above—which employs categories of meaning presumably shared by all Americans—each taste was also coded according to its prevalence in the study population—a dimension of meaning specific to the local context. A continuous measure of popularity is intuitively appealing but difficult to operationalize. Do two tastes that are somewhat popular, for instance, carry the same social meaning as one taste that is very popular? Instead, we follow the assumption of cultural ecologists that there exist "social thresholds" (Kaufman 2004) beyond which a taste will be generally recognized as either popular or unpopular, or in our terminology,

---

sales rather than gross profits), and we do not claim our measures are flawless. Nonetheless, measures were chosen based on a thorough search for alternatives and a conscientious balance between parsimony and theoretical precision. In app. A, we provide additional details on the strengths and limitations of our coding as well as supplementary analyses exploring whether our results might be influenced by differences in operationalizations across domains.

[20] Sales calculations are based on the total number of "units" sold under a given artist's name. The definition of a unit has changed over time, corresponding with changes in available music formats. Today, eligible formats include digital downloads (singles, full albums, kiosks, music videos, mobile phone downloads, and subscription services) and physical purchases (CDs, CD singles, cassettes, LPs/EPs, vinyl singles, music videos, and DVD videos). In 2008, 32% of all music purchased was in a digital format.

"common" or "specialized."[21] While optimal levels for these thresholds would best be determined using qualitative research, we defined a taste as specialized if it was expressed by fewer than .5% of respondents and as common if it was expressed by more than 5% of respondents. These thresholds established categories that were neither too broad (e.g., almost every taste being counted as specialized) nor too restrictive (e.g., almost no tastes being counted as common), produced relatively balanced distributions of specialized and common capital among students, and resulted in classifications that fit our informal understandings of unusual and typical preferences among college students at the time (with some variation particular to this study population). Alternative criteria, such as standard deviations from the mean, did not work due to the heavy skew in taste popularity (i.e., many tastes that few students like and few tastes that many students like). Supplementary analyses (see app. A) suggest results are robust to alternative thresholds set at half and double the above values.

Measures and Mechanisms

Based on the above coding, we created three sets of covariates corresponding to our three theoretical mechanisms. First, to measure dyadic conversion, we calculated the total quantity of consecrated, mass, specialized, and common tastes each pair of students shared in common, as well as the quantity of movie, music, and book tastes each pair of students shared in common. Second, to measure generalized conversion, we calculated the total quantity of consecrated, mass, specialized, and common tastes each student expressed as well as the quantity of movie, music, and book tastes each student expressed.[22] Finally, to measure cultural matching, we calculated the quantity of tastes each pair of students shared in common as a percentage of the total number of unique tastes collectively expressed among the dyad.

Correlations among our 12 original individual variables (4 measures $\times$ 3 domains) are presented in table 3. Aggregated across all waves, the average within-domain correlation was .43 for movies, .47 for music, and .41 for books; the average within-measure correlation was .24 for consecrated culture, .25 for mass culture, .29 for specialized culture, and .39 for common culture; and the average cross-domain, cross-measure correlation was only .17. In other words, students who liked one kind of movie also tended to like

[21] We use "common" instead of "generalized"—the standard antonym of "specialized" in the niche literature (Popielarz and Neal 2007)—to distinguish this term from our generalized conversion mechanism.

[22] It is important to note that our measures of the total quantity of movies, music, and books each student expressed also include tastes that were *not* classified as consecrated, mass, specialized, or common and that the same taste can be classified under multiple codes. Consequently, no linear dependence between taste measures exists by design.

TABLE 3

Correlations between Individual Taste Variables

		Consecrated			Mass			Specialized			Common		
		Movies	Music	Books	Movies	Music	Books	Movies	Music	Books	Movies	Music	Books
Consecrated	Movies. . . .												
	Music	**.20**											
	Books	**.23**	**.30**										
Mass	Movies . . .	*.42*	.07	.15									
	Music	.20	*.47*	.20	**.24**								
	Books	.20	.14	*.48*	**.29**	**.21**							
Specialized	Movies . . .	*.39*	.18	.09	*.43*	.14	.15						
	Music	.10	*.51*	.09	.10	*.39*	.07	**.29**					
	Books	*.22*	.18	*.21*	.23	.13	*.35*	**.35**	**.24**				
Common	Movies . . .	*.45*	.14	.20	*.63*	.34	.32	*.23*	.14	.18			
	Music	.16	*.43*	.25	.19	*.72*	.26	.06	*.32*	.09	**.44**		
	Books	.23	.16	*.54*	.34	.22	*.66*	.11	.05	*.20*	**.41**	**.32**	

Note.—The above correlations refer only to students in the final sample ($N = 520$) and aggregate data from waves 1–3. Bold data are correlations within measures; italic data are correlations within domains.

movies in general (and so on for music and books), and students who liked one kind of movie also tended to like the same kind of music and books—supporting our decision to aggregate tastes by domain and measure.[23]

Relational Data

As DiMaggio has observed, college students "form a kind of ready-made community that is relatively bounded off from the outside world" (2004, p. 101). When collected in such a setting, data on digital interaction can illuminate patterns of relationships among students (Adamic and Adar 2003; Marmaros and Sacerdote 2006; Kossinets and Watts 2009). Facebook is constructed around a formal network of friendships that users can form. To become friends, one user must simply select a link on another user's profile, and a request for friendship is sent. If the other user confirms the friendship, a tie is created; if the user rejects or ignores the request, nothing happens. Once created, a friendship can be dissolved by either user at any time.

The meaning of a Facebook friendship varies across contexts and individuals (e.g., Zywica and Danowski 2008). Students also vary in their tendency to initiate ties with others and to confirm the requests they receive. Informal observation suggests it is common to befriend someone you have met even once, while friendships are rarely created between two people who have never met (Ellison, Steinfield, and Lampe 2007). While it is therefore reasonable to assume the lower limit of tie strength falls somewhere at the "weak tie" or "acquaintance" threshold (Lewis et al. 2008*b*, p. 332; Mayer and Puller 2008, p. 332), Facebook friendships also subsume a variety of other relationships and tie strengths (not unlike our lay definition of "friend"; Fischer 1982). We conservatively interpret these relationships as a generic measure of "friendship" or "social tie."

In addition to ambiguity regarding the meaning of these ties, Facebook friendships are limited in two ways. First, ties are created far more often than they are dissolved (table 2). While this is consistent with our interpretation above—students make many new acquaintances in college, and Facebook itself may help maintain precisely such "weak tie" relationships (Ellison et al. 2007)—social norms against "defriending" also play a role, creating occasional gaps between our instrument and reality. Second, Facebook friend-

[23] An alternative approach would be to retain all 12 original variables in order to examine, e.g., whether the effect of consecrated culture books differs from the effect of consecrated culture music or whether the effect of mass culture movies differs from the effect of common culture movies. Such an exploration of "interactions" between domains and measures not only exceeds the theoretical scope of this article but—based on preliminary analyses—results in drastically reduced statistical power and corresponding challenges for interpretation.

ships are undirected, such that it is impossible to determine who initiated each tie. This is problematic not only because we cannot control for a norm of reciprocity in "accepting" friend requests (cf. Wimmer and Lewis 2010) but also because we cannot determine whether the effects of tastes on network size result from activity or popularity. For instance, our generalized conversion mechanism implies that students with locally esteemed tastes will have larger networks not because they are especially gregarious (outgoing ties) but because they are especially popular (incoming ties). Alternative, directed measures of friendship are necessary to distinguish between these two scenarios.[24]

RESULTS

Results are presented in two stages. In the first stage, we estimate models over all four waves, thereby assuming the parameters governing network evolution are homogeneous over time. While this has the advantage of highlighting the strongest overarching trends based on the aggregated pool of data, temporal differentiation is obscured. In the second stage, we estimate models separately for each transition period—wave 1 to wave 2, wave 2 to wave 3, and wave 3 to wave 4—trading reduced statistical power for greater temporal precision.[25]

Within each stage of results, we develop our models in multiple steps. Model-building with stochastic actor-based models is as much art as science. Unlike mainstream statistical approaches—where deductive hypothesis testing is commonly prioritized over fit—inductive exploration is encouraged in order to find the best-fitting model (Snijders et al. 2010; Ripley et al. 2016; for similar approaches in the networks literature, see Goodreau 2007; Wimmer and Lewis 2010). Results for all periods feature both a "full" model

[24] In fact, our data set contains such a measure: whether one student has uploaded and "tagged" a photo of another student (see Wimmer and Lewis 2010). However, so few students had publicly available photo albums by wave 4 that using this measure would have resulted in a dramatically smaller longitudinal sample.

[25] All models were estimated using RSiena version 1.1-294. We used conditional method of moments estimation with five phase 2 subphases and calculated SEs using the default score function method and 4,000 phase 3 iterations. To parallel the process whereby Facebook friendships are created and dissolved, we used the "initiative/confirmation" model for undirected networks. The t-ratios for all terms in all models were less than 0.1 in absolute value, and the overall maximum convergence ratio for all models was well beneath 0.2, both of which demonstrate excellent model convergence and are the suggested standard for published results. We also found no concerning evidence of multicollinearity: parameter correlations were all less than .9 in absolute value for all models, and parameter and standard error estimates were stable across model runs (Ripley et al. 2016).

including all available covariates as well as a "final," best-fitting model. (The first section of results also includes a control model and a model without foci effects.) In order to reach our final model, we follow a replicable adaptation of the general guidelines suggested by Snijders et al. (2010, pp. 50–51). Specifically, beginning with the full model, we deleted nonsignificant taste effects one by one (in order of lowest-to-highest ratio of parameter estimate to standard error) until only significant taste effects remained—and *unless* the deletion of an effect resulted in another coefficient dropping beneath the threshold of significance (in which case we moved to the next effect). With each successive deletion, we also utilized the score-type goodness-of-fit test described by Schweinberger (2012) to see whether any of the excluded effects should be added back to the model. Because such an approach will be unfamiliar to many readers—and may look suspiciously like "theory trimming" (McPherson 1976)—we provide further context in appendix B along with assessments of fit.

Finally, a brief note on interpretation: as described earlier, the objective function can be used to compare how attractive various tie changes are for a given actor, where the probability of a change is higher as the objective function for that change is higher (subject to the constraints of current network structure as well as random influences). Parameter estimates can therefore be interpreted similarly to those obtained by logistic regression, that is, in terms of the likelihood of idealized micro steps. For instance, a coefficient of .5 for the "shared consecrated culture" effect means that for every additional consecrated culture taste two students share in common, the log-likelihood of friendship formation between these students increases by .5 (all else equal).

Comprehensive Models

Table 4 presents results for our comprehensive models spanning all available data. Model 1 is a control model, containing only nontaste effects. These comprise "structural effects" reflecting the influence of network position on tie formation; "individual effects" reflecting the influence of various individual covariates on sociality, or the propensity to form ties with anyone; and "dyadic effects" reflecting the tendency for a tie to develop between two students who share certain characteristics. (There are also three "rate parameters" indicating the expected number of opportunities every student receives to change a tie during each period.)

The negative, robustly significant "density" coefficient indicates that ties are created relatively infrequently (as opposed to a positive coefficient, which would indicate that ties are more likely to be present than absent). Two students are significantly more likely to become friends the more friends they share (triadic closure), and students are less likely to form new friendships the more friends they already have (preferential attachment). The latter finding is an interesting reversal of the Matthew effect commonly documented in social networks and could be explained in terms of some upper limit on the quantity of contacts a college student can actively maintain (cf. Gonçalves, Perra, and Vespignani 2011).[26] Women form significantly more ties than men; Hispanic students form significantly more ties than white students, while black students form significantly fewer; and students form more ties the higher their zip code tabulation area median household income. Students form significantly fewer friendships the longer they have been members of Facebook. In order of smallest to largest effect, we also find that two students are significantly more likely to become friends if they share the same gender, region of origin, racial background, major, or residence. (Similarity in median household income also fosters tie formation, but the effect is not directly commensurable because the underlying variable is continuous.) Notably, the effect of each additional shared Facebook group on the likelihood of tie formation is as strong as the effect of sharing the same region of origin.

Model 2 is our "full" model—the same as model 1 except that it also incorporates all taste effects suggested by our theoretical framework. In general, the size and significance of control terms is fairly consistent between models. This suggests the effects of network position, demographic traits, online activity, and shared foci cannot be explained by tastes. Beyond these controls, however, we find that cultural tastes independently impact network evolution in a variety of ways. Specifically, students with more mass culture tastes or more tastes in movies are significantly less likely to form ties (generalized inhibition), while students with more common culture tastes are significantly more likely to form ties (generalized conversion). Two students whose overall taste profiles are more similar are also significantly more likely to become friends (cultural matching).

Model 3 is identical to model 2 except that effects for shared foci are omitted. As demonstrated in models 1 and 2, residential facilities, academic majors, and Facebook groups (which we interpret as a proxy for extracurricular activities and a wide array of miscellaneous similarities) powerfully influence network ties by providing opportunities for interaction. If in fact students self-select into these settings on the basis of demographic or cultural traits, then removing these effects should lead to an increase in the absolute

[26] We experimented with five other theoretically plausible structural effects: a square root version of our preferential attachment effect, a degree effect on the rate of change, a degree assortativity effect (both standard and square root versions), and a brokerage effect. In all cases, either the effect was not statistically significant or including the effect resulted in estimation problems (e.g., unsatisfactory convergence, multicollinearity).

TABLE 4
STOCHASTIC ACTOR-BASED MODELS OF CAPITAL CONVERSION, ESTIMATED OVER ALL FOUR WAVES

	Model 1	Model 2	Model 3	Model 4
Rate parameters:				
Period 1	13.92	13.81	14.21	13.82
	(.23)	(.23)	(.24)	(.23)
Period 2	6.21	6.26	6.36	6.26
	(.14)	(.14)	(.15)	(.14)
Period 3	5.12	5.19	5.23	5.19
	(.17)	(.17)	(.17)	(.17)
Structural effects:				
Density	−75.84***	−74.36***	−59.18***	−74.26***
	(3.49)	(3.52)	(3.54)	(3.48)
Triadic closure	14.26***	14.17***	14.65***	14.18***
	(.34)	(.33)	(.33)	(.34)
Preferential attachment	−.45***	−.50***	−.60***	−.51***
	(.07)	(.07)	(.07)	(.07)
Individual effects:				
Demographics:				
Female	9.43***	6.97*	5.43*	6.81*
	(2.67)	(2.94)	(2.72)	(2.77)
Black[a]	−24.61***	−19.87***	−15.08**	−19.77***
	(5.18)	(5.29)	(5.24)	(5.28)
Asian[a]	−1.11	−2.22	.63	−2.06
	(3.96)	(3.99)	(4.04)	(4.02)
Mixed[a]	12.01	4.74	3.25	5.45
	(7.72)	(7.81)	(7.75)	(7.49)
Hispanic[a]	12.39*	13.69*	18.01***	14.29**
	(5.28)	(5.36)	(5.15)	(5.37)
Median household income (K)	.18***	.18***	.14**	.17**
	(.05)	(.05)	(.05)	(.06)
Online activity:				
Days since last update	−.09	−.10*	−.11*	−.10
	(.05)	(.05)	(.05)	(.06)
Days since joined Facebook	−.13**	−.14***	−.11***	−.14***
	(.04)	(.04)	(.03)	(.04)
Tastes:				
Consecrated culture		−.61	−1.02*	−.28
		(.43)	(.43)	(.40)
Mass culture		−.81*	−.44	−.72*
		(.34)	(.34)	(.32)
Specialized culture		−.01	−.27	
		(.37)	(.35)	
Common culture		2.15***	1.92***	2.25***
		(.52)	(.50)	(.37)
Movies		−1.06***	−.76*	−1.06***
		(.31)	(.30)	(.22)
Music		.17	.22	
		(.25)	(.24)	
Books		.73	.74	.69*
		(.41)	(.39)	(.34)

TABLE 4 (*Continued*)

	Model 1	Model 2	Model 3	Model 4
Dyadic effects:				
Shared demographics:				
Same gender	8.41***	8.04***	9.86***	8.11***
	(1.84)	(1.86)	(1.81)	(1.88)
Same racial background	30.48***	29.67***	31.37***	29.61***
	(2.55)	(2.52)	(2.48)	(2.54)
Similar median household income (K)	25.38**	23.07**	26.07**	22.97**
	(7.95)	(7.91)	(7.93)	(7.98)
Same region of origin	9.38***	9.58***	10.83***	9.60***
	(2.68)	(2.64)	(2.64)	(2.67)
Shared foci:				
Shared residence	100.84***	100.93***		100.95***
	(2.37)	(2.45)		(2.40)
Shared major	55.43***	55.89***		55.86***
	(3.35)	(3.41)		(3.46)
Shared Facebook group	9.43***	9.45***		9.44***
	(.81)	(.80)		(.81)
Shared tastes:				
Shared consecrated culture		3.35	2.87	
		(2.41)	(2.40)	
Shared mass culture		−3.41	−3.11	−2.97
		(1.90)	(1.83)	(1.78)
Shared specialized culture		14.07	13.18	16.37*
		(8.22)	(8.16)	(8.16)
Shared common culture		−3.04	−3.26	−2.09
		(2.11)	(2.06)	(1.68)
Shared movies		1.85	3.14	
		(2.45)	(2.45)	
Shared music		2.81	3.45	2.58*
		(1.87)	(1.78)	(1.31)
Shared books		.81	2.17	
		(2.89)	(2.88)	
Taste similarity (%)		1.94*	1.91*	2.15***
		(.77)	(.75)	(.63)

NOTE.—Significance levels are not provided for rate parameters, as testing that they are zero is meaningless (if they were zero, there would be no change between successive waves). N = 520 for all models. All results (except rate parameters) are multiplied by 100 to preserve space. Data are presented as coefficient (SE).

[a] Dummy-coded variables, with "white" as reference category.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

value of coefficients representing such traits (because their effects on tie formation would no longer be mediated). Interestingly, this is only selectively the case. All four effects for shared demographics indeed increase slightly from model 2 to model 3—an unsurprising finding insofar as many extracurricular activities and academic majors are more attractive to students from particular backgrounds. Individual effects of tastes on ties are fairly consistent across models, however, and the only effect that becomes significant is the (negative) effect of consecrated culture. Meanwhile, although multiple parameter estimates for shared tastes increase—especially the three domain measures (shared movies, music, and books)—these effects do not cross the threshold of significance.

We then pursued the model development approach described above. This resulted in model 4, our final model. As required by our approach, the same taste effects that were significant in model 2 continue to be significant in model 4. Differences in the sizes of these parameter estimates are negligible. However, by eliminating those effects that do not contribute to model fit, we find three new significant effects that were masked in model 2: the more tastes in books a student expresses, the more friendships she is likely to form; the more specialized culture tastes two students share, they more likely they are to become friends; and the more tastes in music two students share, the more likely they are to become friends. The size of the dyadic conversion effect for specialized culture (β = .164) is particularly noteworthy: students who share a single specialized culture taste are, ceteris paribus, more likely to become friends than students who share the same gender (β = .081) or region of origin (β = .096); and students who share just two specialized culture tastes are more likely to become friends than students who share the same racial background (β = .296). Also included are the individual effect for consecrated culture and the dyadic effects for shared mass culture and common culture. While these effects are not themselves statistically significant, their omission results in the dyadic effect for shared music dropping beneath significance, and so we maintained them per our model-building protocol. In other words, sharing tastes in music is significantly related to tie formation, but only when we also control for consecrated culture tastes and shared mass and common culture tastes.

Period Models

We next estimated separate models for each transition period—first, to assess whether any of the effects in our comprehensive models may have been produced by atypical behavior during a single period (cf. Lewis et al. 2012) and second, because our theoretical framework leads us to consider whether or how capital conversion dynamics vary over time as an important but seldom explored question in its own right. In table 5, we replicate the above presentation of results but focus exclusively on the full and final models for the sake of concision.

Two prefatory comments are in order. First, it is important to remember that each model conditions on its starting point, and it is only the transition between waves that is considered. So, for instance, the models for period 2

TABLE 5

Stochastic Actor-Based Models of Capital Conversion, Estimated Separately for Each Period

	Period 1		Period 2		Period 3	
	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Rate parameter	13.08	13.07	6.51	6.51	5.57	5.56
	(.22)	(.21)	(.15)	(.15)	(.18)	(.18)
Structural effects:						
Density	−77.09***	−77.41***	−45.18***	−45.66***	−103.69***	−102.30***
	(4.45)	(4.48)	(7.49)	(7.19)	(11.09)	(10.49)
Triadic closure	15.04***	15.04***	14.00***	14.02***	13.56***	13.64***
	(.49)	(.49)	(.56)	(.54)	(.87)	(.80)
Preferential attachment	−.36***	−.35***	−.94***	−.93***	−.35	−.38*
	(.09)	(.09)	(.14)	(.13)	(.20)	(.18)
Individual effects:						
Demographics:						
Female	3.63	4.76	11.87*	12.51*	12.19	11.36
	(3.87)	(3.57)	(5.30)	(5.30)	(7.47)	(7.06)
Black[a]	−2.69	−1.61	−64.41***	−63.24***	−36.41*	−38.84**
	(6.80)	(7.10)	(10.64)	(10.74)	(15.33)	(14.94)
Asian[a]	4.45	4.79	−10.09	−9.81	−25.54*	−24.84*
	(5.37)	(5.59)	(7.36)	(7.46)	(12.13)	(11.40)
Mixed[a]	17.57	17.80	−43.52**	−43.09**	45.98*	44.16*
	(10.61)	(10.57)	(15.84)	(16.02)	(21.18)	(19.37)
Hispanic[a]	39.66***	40.55***	−24.46*	−23.74*	−33.61*	−34.93*
	(7.17)	(7.16)	(10.86)	(10.80)	(14.77)	(14.43)
Median household income (K)	.39***	.40***	−.09	−.08	−.12	−.13
	(.07)	(.07)	(.11)	(.11)	(.15)	(.16)

TABLE 5 (*Continued*)

	Period 1		Period 2		Period 3	
	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Online activity:						
Days since last update	−.13	−.14*	.02	.02	−.16	−.17
	(.07)	(.07)	(.10)	(.10)	(.15)	(.14)
Days since joined Facebook	−.18***	−.19***	.00	.00	−.26**	−.26**
	(.05)	(.05)	(.06)	(.06)	(.10)	(.10)
Tastes:						
Consecrated culture	−.83		−1.08	−.99	.96	
	(.56)		(.82)	(.77)	(1.12)	
Mass culture	−.65	−.80*	−.64	−.51	−3.50**	−3.67***
	(.41)	(.38)	(.74)	(.67)	(1.17)	(1.04)
Specialized culture	−.21		−.74		.39	
	(.49)		(.73)		(1.03)	
Common culture	1.61*	1.85***	−.68		5.68**	5.91***
	(.69)	(.45)	(1.16)		(1.77)	(1.32)
Movies	−1.01*	−1.01***	−.49	−.98*	−.78	
	(.40)	(.28)	(.61)	(.47)	(.97)	
Music	.11		.92	.60*	.28	.76
	(.33)		(.51)	(.24)	(.70)	(.39)
Books	.84		1.64*	1.35*	.29	
	(.52)		(.81)	(.67)	(1.19)	
Dyadic effects:						
Shared demographics:						
Same gender	10.69***	10.60***	4.45	4.27	6.15	6.07
	(2.45)	(2.49)	(3.50)	(3.44)	(5.02)	(5.07)
Same racial background	35.97***	35.96***	12.87**	13.08**	29.06***	29.37***
	(3.25)	(3.20)	(4.94)	(4.94)	(7.43)	(7.20)
Similar median household income (*K*)	22.37*	22.86*	24.09	25.04	33.75	32.17
	(10.59)	(10.26)	(15.60)	(15.59)	(23.42)	(23.50)
Same region of origin	4.40	4.40	15.67**	15.68**	16.76*	16.72*
	(3.56)	(3.45)	(4.96)	(5.02)	(7.07)	(7.38)

Shared foci:						
Shared residence	101.75***	101.82***	100.70***	100.81***	103.08***	102.92***
	(3.23)	(3.14)	(4.62)	(4.42)	(6.44)	(6.58)
Shared major	54.17***	54.18***	63.24***	63.28***	49.28***	49.59***
	(4.54)	(4.49)	(6.33)	(6.38)	(10.10)	(9.70)
Shared Facebook group	12.05***	12.15***	5.70***	5.75***	5.90**	5.81**
	(1.05)	(1.06)	(1.58)	(1.58)	(2.26)	(2.25)
Shared tastes:						
Shared consecrated culture	4.46		−.44		4.04	
	(2.95)		(4.99)		(7.52)	
Shared mass culture	−2.13		−8.77*	−7.40*	−2.52	
	(2.21)		(4.23)	(3.28)	(7.15)	
Shared specialized culture	17.67	20.24*	9.70		−5.74	
	(10.33)	(9.88)	(17.13)		(30.12)	
Shared common culture	−3.16	−2.53	−.13		−3.48	
	(2.52)	(1.73)	(4.57)		(7.68)	
Shared movies	3.06		−3.78		8.09	
	(3.10)		(5.20)		(8.93)	
Shared music	1.65		4.85		9.33	
	(2.27)		(3.77)		(7.09)	
Shared books	−1.74		3.89		9.67	
	(3.70)		(5.65)		(9.99)	
Taste similarity (%)	2.62**	3.04***	1.82	2.51**	−2.82	
	(.99)	(.80)	(1.28)	(.92)	(2.99)	

NOTE.—Significance levels are not provided for rate parameters, as testing that they are zero is meaningless. $N = 520$ for all models. All results (except rate parameters) are multiplied by 100. Data are presented as coefficient (SE).

[a] Dummy-coded variables, with "white" as reference category.

* $P < .05$.

** $P < .01$.

*** $P < .001$.

take for granted the state of the network at March of the students' sophomore year and parameter estimates refer primarily to the new ties students formed over the succeeding 12 months.[27] Second, it is important to note that missing network data increased over time (table 2) and were particularly prevalent at wave 4 (students' senior year). While our estimation technique treats missing data as noninformative (Huisman and Steglich 2008; Ripley et al. 2016),[28] parameter estimates for period 3 may be especially imprecise; interpretation should be correspondingly tentative.[29]

We begin with changes in control effects over time. Several trends are noteworthy. Triadic closure and living in the same residence are the two most robust and consistent predictors of tie formation: over all three periods, student friendship patterns are powerfully influenced by both physical and "relational" opportunity structures (cf. Kossinets and Watts 2009). Our measures of online activity, recorded at wave 1, display inconsistent effects over time. Interestingly, many of the demographic effects on sociality we observed in table 4 are strongest at periods 2 and 3. The effects of shared gender, racial background, and Facebook groups, in contrast, are strongest at period 1, although the latter two remain statistically significant at all periods. Region of origin—an attribute that might be "invisible" at first outside of dialect or dress—becomes much more important to tie formation over time. Trends in socioeconomic homophily are difficult to interpret due to large standard errors. Finally, sharing the same major is even more important for friendship formation across all three periods than sharing the same racial background. This effect is strongest at period 2 and weakest at period 3—perhaps because students take more classes for their major over time (as opposed to general education requirements), but by period 3 most students in the same major who will become friends already have become friends.

[27] We say "primarily" rather than "exclusively" because these estimates are derived not just from students' tendency to form new ties but also from their tendency to maintain (rather than dissolve) existing ties. However, because tie dissolution is relatively rare, patterns of dissolution contribute proportionately little to the coefficients we report.

[28] Specifically, during the simulation phase of estimation, missing values are imputed to allow for meaningful simulations: missing network variables are replaced by their last observed value (and a "nontie" in the case of no earlier observation), and missing covariates are replaced by the variable's global mean. For the calculation of target statistics used by the method of moments, however, only nonmissing data are used (Ripley et al. 2016, pp. 32–33). Levels of missing data (individual and relational) in this study fall well within recommended thresholds, with the sole exception of network data at wave 4. Even there, however, the large size of our network may mitigate this concern; potentially more worrisome is that these data are unlikely to be missing at random.

[29] An alternative approach would be to maximize available data for each transition period separately, omitting missing data altogether. However, this would create a situation where the study sample varies among models, so it would be unclear whether differences in results should be attributed to differences in network dynamics or differences in sample composition over time.

Shifting attention to the focal taste effects of this article, models 5–10 (table 5) clarify and expand upon the time-invariant results from model 4. Most significant taste effects from model 4 (the comprehensive final model) are significant in model 6 (the period 1 final model), including the generalized inhibition effects of mass culture and movies, the generalized conversion effect of common culture, the dyadic conversion effect of specialized culture, and cultural matching. Interestingly, however, only two of these effects (matching and the inhibition effect of movie tastes) are still significant in period 2 (model 8)—where tastes in music and (especially) in books are now also a resource for generalized conversion. There is also an unexpected negative and significant dyadic effect of sharing mass culture tastes. Finally, in period 3, shared tastes do not significantly contribute to friendship formation, whether through dyadic conversion or cultural matching (model 10). However, our model-building approach in this instance is potentially misleading: unusually large standard errors obscure several unusually large parameter estimates in model 9—especially for shared movies, music, and books. Common culture tastes are a particularly strong predictor of generalized tie formation in period 3, while the more mass culture tastes a student expresses, the fewer ties she forms between her junior and senior years.[30]

DISCUSSION

It is important to remember that Facebook friends are, at best, imperfect measures of friendship; that tastes and conversion dynamics among this sample may not be representative of other people in other contexts; and that missing data problems—owing to both privacy settings and nonreport—prevent us from studying all students and detract from the precision of our estimates. Nonetheless, the above results powerfully illustrate the utility of our theoretical framework and suggest both qualifications and extensions to prior work.

While research on tastes has overwhelmingly relied on exogenous measures, we find that both exogenous and endogenous properties of tastes have consequences for network dynamics—each controlling for the other. In addition, we provide robust support for entirely new mechanisms of capital conversion that naturally extend prior work: just as students from particular demographic backgrounds are more or less likely to form ties (Goodreau

[30] To confirm these results are not an artifact of our method—or our approach to model building—we replicated our four final models using alternative, more basic techniques: first, a descriptive comparison of dyads that did and did not transition to friendship; and second, multivariate logistic regression. While the assumptions of logistic regression conflict with the nature of our data (specifically, our data are not a random sample and observations are clearly interdependent)—and indeed, stochastic actor-based models were designed to surmount many of the limitations of alternative methods (Steglich et al. 2010)—results were overwhelmingly consistent with those reported above and are presented in a detailed online supplement for the interested reader.

et al. 2009; Wimmer and Lewis 2010), so some tastes (e.g., for common culture and for books) constitute a form of "generalized capital" for creating friendships with anyone. Other tastes, meanwhile (e.g., for mass culture and for movies), have a negative effect on generalized tie formation and thus inhibit the creation of social capital. And while prior work has relied on cultural homophily as a central theoretical assumption (Carley 1991; Axelrod 1997) or else examined this mechanism qualitatively (Rivera 2012), we provide quantitative evidence for cultural matching above and beyond the effects of dyadic conversion: students are attuned to the overall similarity of cultural profiles, suggesting that tastes as well as nontastes are important for forming friendships (cf. Edelmann and Vaisey 2014).

We paused to explicitly address the question of whether tastes impact ties indirectly by leading certain kinds of students to self-select into certain kinds of meeting opportunities (Feld 1981; Benediktsson 2012). While residence assignments are randomly determined, this could occur for academic majors and many kinds of student organizations. We expected that if the impact of some tastes is absorbed by social foci, removal of foci effects should elevate taste coefficients accordingly. Surprisingly, this was hardly the case: taste coefficients changed little between models 2 and 3, implying tastes' impact on network evolution is primarily direct rather than mediated. Two exceptions were that the parameter estimates for shared movies, music, and books increased more than other coefficients, although not to the point of significance, and that the (negative) effect of consecrated culture on generalized tie formation became significant in model 3. The latter suggests that one reason students with consecrated culture tastes tend to have smaller networks might be because they dedicate time to "greedy" organizations (Coser 1974). In sum, social foci play an undeniably important role in network evolution. However, they do not seem to be central to capital conversion—at least for the tastes, foci, and students examined here.[31]

We also provide, to our knowledge, the first direct comparison of how the social consequences of tastes vary by domain. In short, prior findings based on music may not be generalizable to other media, and different kinds of

[31] Because our models condition on the state of the network at wave 1, one explanation is that tastes' indirect effect on ties (via foci) was effectively conditioned out of our analysis. In other words, if students with particularly strong cultural interests had already joined relevant groups (and become friends with most people in these groups) by the time of our first data pull (i.e., prior to March of their freshman year), our models would be unable to discern these effects. The fact that the impact of Facebook group comembership on tie formation is strongest at period 1 would seem to support this explanation. An additional limitation is that we are here forced to operationalize social foci as yet another shared trait that may increase or decrease the probability of tie formation, while in reality, foci are meeting structures that may shape students' very awareness of friendship opportunities. Realistically conceiving the relationship between physical and relational space is a challenging methodological puzzle and an area of ongoing development across social network methods at large (see adams, Faust, and Lovasi 2012).

tastes provide different paths to friendship. Further, the salience of these paths varies depending on the time horizon at hand. Here, perhaps, is our most striking result: we find conversion dynamics vary not only by how we measure tastes and what we mean by "conversion," but by when we collect our data. Building on prior distinctions between meeting and mating (Verbrugge 1977), visible and invisible attributes (de Klepper et al. 2010), and varying levels of network constraint (Feld 1981), we suggest different kinds of tastes are salient at different points in time in relatively close-knit settings (such as a residential college campus where students eat, sleep, and study together as well as take classes and socialize). Our results speak to three temporal stages corresponding to the three periods we examined.

In the early stages of network formation—when students are just becoming acquainted—most tastes are indeed invisible, and the crucial question is what stands out. Perhaps unsurprisingly, it is here when shared tastes that are particularly unusual are particularly conducive to friendship formation, that is, specialized culture dyadic conversion (cf. Adamic and Adar 2003; Lewis et al. 2012). If students share so much in common that their overall taste profiles are similar, this is also unlikely to escape recognition, leading to, cultural matching. Both dynamics may fade over time as friendships are less influenced by first impressions.

We also observe three generalized effects of tastes at period 1. First, students with more mass culture tastes form significantly fewer friendships. Particularly in an elite setting, students who openly express (on their Facebook pages) tastes that follow public opinion at large may be viewed as lacking in cultural refinement—and avoided accordingly (cf. Lopes 2006). This makes the second generalized effect especially interesting: controlling for exogenous popularity, expressing tastes that are locally popular is instead rewarded, as students with more common culture tastes form significantly more friendships overall. Third, we find a generalized inhibition effect for tastes in movies. We return to these latter two effects below.

As students settle into their residential communities, alternative forms of capital come to the fore. Although slightly diminished in effect size, matching continues to have a robust impact on network evolution at period 2. Unique to this period, however, are strikingly divergent effects of different domains: students who express more tastes in music and (especially) books form more ties overall, while students who express more tastes in movies again form fewer. We speculate that the generalized conversion effects of music and books are driven by distinct social signals conveyed by each medium. Given that domain effects persist net of endogenous and exogenous measures, these students may be "omnivores by volume" who "engage in more activities and express preferences for a wider range of cultural items than others" (Warde and Gayo-Cal 2009, p. 121). Insofar as omnivorousness in music is a marker of high status, it is no surprise that these students are popular in an elite institution. However, particularly in a college setting—where education is universally valued and reading is a core component of learning (see Bennett et al. 2009)—bookworms display a different kind of locally situated capital that others may wish to benefit from.[32] Tastes in movies, meanwhile, represent the opposite scenario: net of cultural measures, movies per se may have neither symbolic nor instrumental value—and might here be viewed as a "base" form of consumption (cf. Chan and Goldthorpe 2005; Bennett et al. 2009).

Surprisingly, we also find at period 2 a significant negative effect for sharing mass culture tastes, indicating students who share such tastes start avoiding each other. Why this occurs is difficult to determine. On an abstract level, it flies in the face of the universal principle that similarity breeds attraction (McPherson et al. 2001) and suggests dyadic aversion may in rare cases accompany dyadic conversion. On a substantive level, the underlying dynamic may be diametrically opposite to the effect of specialized culture at wave 1: just as sharing locally unusual tastes creates a powerful social bond—the fact that we are similarly different from them fosters a special affinity between us—sharing universally popular tastes may be uniquely socially repelling—the thing we have in common is our ordinariness vis-à-vis the average American.

Finally, between students' junior and senior years (period 3), most students who will become friends have become friends already. Missing network data at wave 4 also result in large standard errors, complicating interpretation. Nonetheless, important trends are evident. First, while not statistically significant, parameter estimates for shared movies, music, and books are higher than for any other period, suggesting ongoing (but imprecisely measured) dyadic conversion. Music tastes also have generalized conversion value (this term is borderline significant, and a score-type test suggested it should be included). Tastes in common culture are strongly associated with forming ties (an effect that was also significant, but much smaller, at period 1), while tastes in mass culture negatively impact tie formation. It seems, therefore, DiMaggio was correct that "popular culture provides the stuff of everyday sociability" (1987, p. 444)—with two qualifications. First, what matters is that the taste is locally popular, not universally popular. Second, people do not need to like the same culture to discuss it (otherwise, we would see

[32] An alternative explanation draws on the "more-more" principle discussed by Robinson and Godbey (1997). This principle—which reflects a Newtonian model of human behavior that "bodies in motion stay in motion"—would suggest some underlying disposition may be jointly responsible for elevated tie formation, cultural consumption, and Facebook participation alike. Insofar as some students are generally more active people, however, we expect that this will be captured by our controls for online activity and preferential attachment; this principle also cannot explain why the association between tastes and network size varies across domains (and is negative for movies).

evidence of dyadic conversion). Rather, culture many students like is probably culture most students know about, so those with locally common preferences can strike up a conversation with just about anyone.

We recognize the dangers of generalizing from a single case. Nonetheless, careful consideration of how the particularities of our context may have influenced results can shed light on broader principles. Inspired by DiMaggio (1987), we thus conclude our discussion with the following propositions about capital conversion—intended both to summarize our findings in general language and to provide a starting point for further exploration:

PROPOSITION 1.—*In a nascent social network, the people most likely to quickly become friends are those whose cultural resemblance is subjectively striking—namely, people who share tastes* and *nontastes across multiple domains and people who share tastes that are locally uncommon.*

PROPOSITION 2.—*Culture many people like is culture most people know about. Consequently, tastes that are relatively common in a particular setting constitute a broad social resource that facilitates friendship with almost anyone—even, and especially, in later stages of network evolution.*

PROPOSITION 3.—*Networks follow culture based on context-specific hierarchies of evaluation. In other words, people who express tastes that are locally valued will be attractive as potential friends and people who express tastes that are locally disvalued will be avoided.*

PROPOSITION 4.—*People who share locally stigmatized tastes will be unattractive to one another as potential friends, especially at intermediate stages of network evolution (i.e., after enough time to learn each other's dirty secrets but before enough time to stop feeling threatened by them).*

PROPOSITION 5.—*Just as unexpected similarities (e.g., sharing uncommon tastes) are most salient as networks emerge, mundane similarities (i.e., sharing* any *tastes) become more salient as networks evolve. Domains more associated with visible identity markers matter sooner.*

PROPOSITION 6.—*Cultural organizations are efficient mediators of tastes and friendship. In other words, people who self-select into shared foci on the basis of their tastes become friends rapidly or not at all; beyond the earliest stages of network evolution, the impact of tastes on ties is largely direct.*

CONCLUSION

In the context of a growing literature on culture and social networks (Pachucki and Breiger 2010), we have focused here on the question of how cultural tastes are converted into social network ties. While this question is rooted in Bourdieu's (1986) classic essay, it was not fully articulated until 20 years later (Lizardo 2006)—and is now benefitting from a flurry of scientific inquiry (Selfhout et al. 2009; Vaisey and Lizardo 2010; Lewis et al. 2012; Edelmann and Vaisey 2014). Nonetheless, as befits any subfield in its infancy, proof of concept has preceded elaboration, and this nascent body of scholarship has yet to be integrated under a more comprehensive theoretical umbrella. This can also be attributed to the practical constraints of available data and methods: there are only so many questions that cross-sectional, closed-ended data on tastes and egocentric data on ties can answer, and the independence assumptions of mainstream statistical methods have always been a poor fit for analyzing networks.

In this article, we have outlined such a broadened framework and illustrated its utility using a unique, longitudinal data set and recent developments in network modeling. In this way, we have not only integrated prior work and empirically demonstrated that tastes impact ties in a variety of unacknowledged ways, we have also attempted to narrow the gap between theory, methods, and data by pushing each to the next level. First, we suggest that formal properties of cultural ecologies can shed light on locally constituted meaning and action net of the effects of traditional measures (Mark 1998; Mohr 1998; Harrington and Fine 2006). Second, we caution against assumptions of time homogeneity and domain consistency in capital conversion dynamics and encourage longitudinal data collection across multiple fields of preferences (Bennett et al. 2009). Third, we recommend the use of stochastic actor-based models for future research on culture and networks—for, as Snijders (2011, p. 146) has suggested, their emphasis on context-dependent choices made by actors provides one possible avenue for integrating structure and agency under a powerful yet flexible quantitative framework (Emirbayer and Goodwin 1994).

This analysis is limited in many ways. Most importantly, our data are restricted to only a subset of students at a single college. While students who are especially culturally and socially active on Facebook present a rich opportunity for examining capital conversion, left unanswered is the question of generalizability to other people in other places and at different stages of the life course. Unfortunately, surveying a random sample of respondents precludes the in-depth measurement of local ecologies and restricts attention to egocentric network data. Such data cannot produce meaningful conclusions about network evolution because crucial data on "nonchoices" are omitted (Steglich et al. 2010). Fortunately, recent developments in meta-analyses for stochastic actor-based models provide a possible solution to this catch-22 (Ripley et al. 2016)—provided that longitudinal, sociocentric network data are collected across multiple settings. While this is no trivial task, the payoff for scholars of culture and networks could be well worth the cost: a comparative approach would illuminate contextual variation in conversion dynamics that is bracketed in the preceding analyses; longitudinal and less ambiguous data on social foci could also be incorporated in a variety of ways to explore the mutually constitutive nature of persons and cultural groups (Breiger 1974; Block and Grund 2014).

In addition to better data and a broader, comparative approach, future research could contribute further nuance to our proposed theoretical framework. With respect to cultural tastes, we have here focused on preferences for movies, music, and books. Beyond the need to examine other domains,[33] prior work has also operationalized tastes as knowledge or consumption (cf. Warde and Gayo-Cal 2009). Tastes that are "weaker" than favorites—or even dislikes—may be equally consequential for networks (Bryson 1996; Schultz and Breiger 2010). Beyond consecrated, mass, specialized, and common, other exogenous and endogenous measures are certainly possible (cf. Peterson 1992; Lewis et al. 2008*a*), and our proposed set of mechanisms could become even more variegated (e.g., by considering dyadic aversion or intertaste compatibility).

With respect to social ties, Lizardo (2006) has already elaborated one way capital conversion might vary by tie strength (see also Edelmann and Vaisey 2014). This is an important omission from our framework, and given that close friends tend to constitute a small portion of Facebook friends (Arnaboldi et al. 2013), our results are likely biased toward weak tie dynamics. (Modeling tie strength longitudinally also raises new methodological challenges insofar as strong ties tend to evolve from weaker relationships.) We further suggest future research move beyond the trinary operationalization of ties as weak/strong/absent and incorporate continuous measures, consider how capital conversion might vary across different kinds of relationships (e.g., who you like, who you know, and who you actually spend time with), and examine the differential role tastes play in tie formation compared to tie maintenance (cf. Selfhout et al. 2009).[34]

A final extension of our approach is worth suggesting. Although we have here focused on the substantive signal conveyed by preferences for different media—for example, how books are different from music and the distinctive social meaning of tastes for each—our emphasis on local ecologies could be extended to understand why capital conversion might vary across domains. In this way, the statistics in table 2 concerning the structure of each "taste field" could become explanatory principles in their own right. For

[33] The Glasgow Teenage Friends and Lifestyle Study, which in addition to data on music genres contains data on the frequency at which students participated in 15 different leisure activities, could be an especially useful starting point (see https://www.stats.ox.ac.uk/~snijders/siena/Glasgow_data.htm).

[34] This is another important limitation of our article, given that friendships on Facebook operate "opposite" to real life: on Facebook, a tie persists by default and requires active intervention to terminate, whereas actual social relationships tend to require ongoing renewal. While RSiena is capable of distinguishing these two processes, we were unable to pursue this possibility. Specifically, "if the network dynamics in a given data set is such that ties mainly are created, and they are dissolved rather rarely, then the data will contain little information about the question whether creating ties follows different rules than dissolving ties" (Ripley et al. 2016, p. 37).

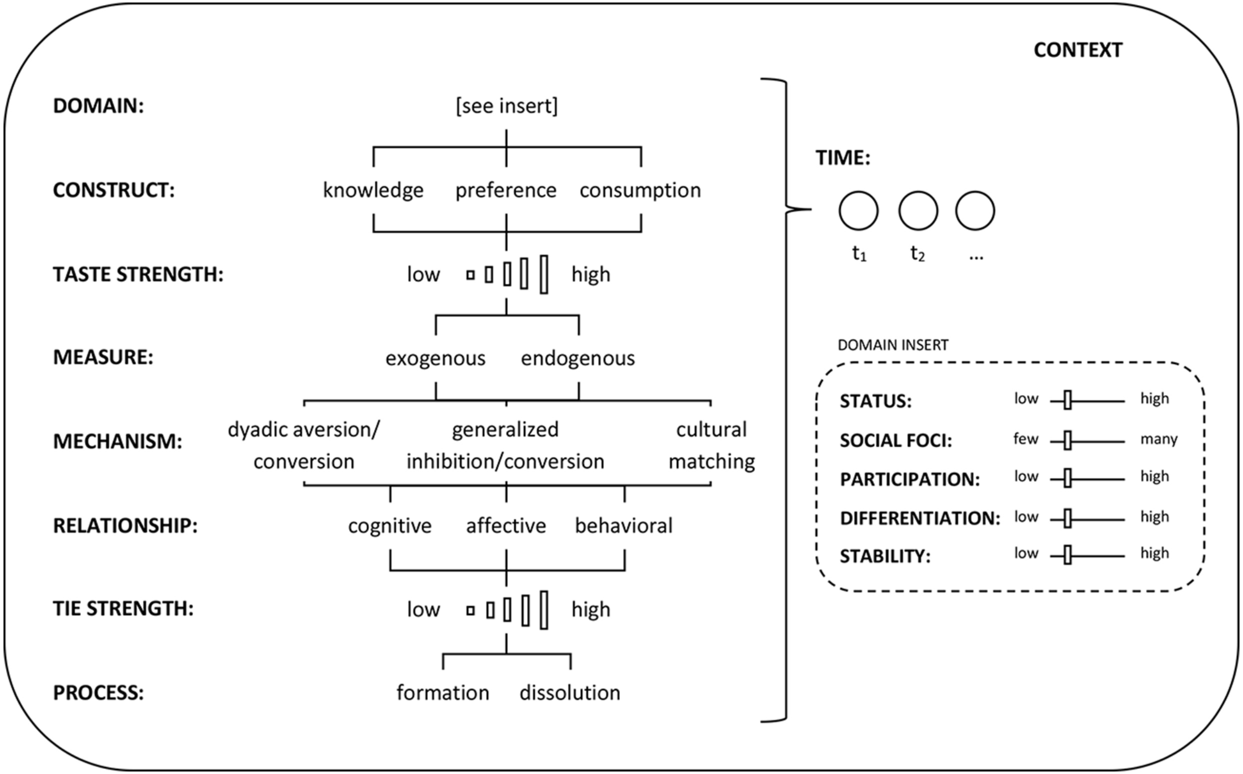


FIG. 3.—Extended capital conversion framework

instance, the fewer people who participate in a given domain, the less the content of those tastes might matter, and it could be the distinction between those who have and do not have preferences that is socially salient (e.g., Bukodi 2007). Differentiated fields may be less hierarchical, facilitating ties across boundaries (DiMaggio 1987), and stable fields may be more important to identity and therefore positively correlated with impact. In this way, even identical tastes with identical popularity may have very different consequences depending on the structure of the surrounding ecology (Kaufman 2004).[35]

We visualize in figure 3 what such an extended version of our framework might look like. It is essential to remember, however, that even this framework is but one part of a much larger story. Just as tastes influence ties, so are preferences transmitted through relationships in ways we are only beginning to appreciate (Steglich et al. 2006; Aral and Walker 2012; Lewis et al. 2012),[36] and both social structure and the "cultural surface" evolve according to their own internal dynamics (Lieberson 2000). We hope future work picks up where we leave off.

APPENDIX A

Coding Cultural Tastes

As a growing proportion of social interaction occurs online, data from digital sources have the potential to address important social scientific questions. Sociologists are increasingly making use of these opportunities (Golder and Macy 2014). Elsewhere, we have provided a detailed introduction to the Facebook data set used in this article and a thorough discussion of its strengths and limitations (Lewis et al. 2008*b*). In particular, our data set combines sociocentric network data with open-ended taste data—both of which

[35] Residential colleges provide one opportunity for exploring these dynamics. Insofar as students at the same college share a common cultural context but still primarily form ties within (randomly assigned) subcommunities like dorm buildings, one could exploit naturally occurring variation in the structure of taste fields across subcommunities to disentangle the effects of local ecology "controlling for" context.

[36] While stochastic actor-based models are capable of disentangling selection and influence, current software cannot implement the kinds of dynamics our framework requires. Specifically, dyadic dependent variables such as taste sharing are not yet available, and the evolution of cultural covariates would need to be systematically integrated due to the fact that these measures are collectively dependent on thousands of underlying indicator variables representing each student's constellation of tastes. Recent advances in the coevolution of one- and two-mode networks provide an exciting alternative approach (e.g., Lomi and Stadtfeld 2014). Instead, we are forced to here "artificially freeze" the values of taste variables at the last preceding observation. While Mercken et al. (2010, fig. 1) give one example of how this may result in a misdiagnosis of social selection when peer influence is actually at work, it is unlikely that such a roundabout chain of events could happen frequently enough to influence our central findings.

are available longitudinally on a standardized template. Drawbacks, meanwhile, include participation biases, incomplete (or out of date) information, and a variety of challenges of interpretation (cf. Lewis 2015*b*).

Here we focus on the issue of coding cultural tastes. While it is possible to study the correlation between friendships and tastes without imposing any structure on the raw taste data (as in Lewis et al. 2008*b*)—and also to group tastes inductively using clustering algorithms (e.g., Wimmer and Lewis 2010; Lewis et al. 2012)—our aim of developing the distinction between exogenous and endogenous meanings required us to classify individual cultural objects using categories that are commensurable with prior literature. Although our choice of measures emerged from a careful consideration of many alternatives, no measure is perfect, and we will be the first to acknowledge residual concerns. Two such concerns are that our exogenous codes capture different distinctions across domains and that our endogenous thresholds are potentially arbitrary.

Exogenous Codes

Among many possible exogenous meanings, consecrated culture and mass culture are attractive categories both because of the attention they have received in prior literature (e.g., Peterson 1992; Allen and Lincoln 2004) and because they lend themselves straightforwardly to objective empirical measures. The choices of which consecration projects to focus on and how to assess mass market success were still open, however, and while differences in operationalization across domains are inevitable (given differences in market structure, evaluation standards, and inherent qualities of movies, music, and books as cultural products), it is important that our various codes for consecrated and mass culture approximate the same underlying concepts.

One way of evaluating whether we achieved this goal is to examine correlations among the various exogenous and endogenous codes for each domain and wave. These correlations are reported in table A1. In general, correlation coefficients are highly consistent within domains and across waves. For instance, the correlation between movies coded as specialized culture and consecrated culture at wave 1 (−.20) is almost identical to the same correlation at wave 2 (−.21) and wave 3 (−.19). Across domains and waves, correlations between the same two codes (e.g., specialized culture and mass culture) also always have the same sign (positive or negative). Finally, while occasionally there is some degree of variation by domain (e.g., the correlation between common culture and consecrated culture is as low as .03 for music but as high as .18 for books), in general we found no evidence of consistent or extreme discrepancies.

A second way of assessing this concern is to ask how this article’s central results might have been different if our measures had focused on a single

TABLE A1
CORRELATIONS BETWEEN TASTE CODES

		Wave 1				Wave 2				Wave 3			
		Consecrated	Mass	Specialized	Common	Consecrated	Mass	Specialized	Common	Consecrated	Mass	Specialized	Common
Movies	Consecrated												
	Mass	.07				.07				.06			
	Specialized	−.20	−.26			−.21	−.28			−.19	−.29		
	Common	.10	.13	−.27		.11	.13	−.24		.12	.16	−.23	
	N		1,918				1,717				1,577		
Music	Consecrated												
	Mass	.11				.11				.12			
	Specialized	−.23	−.35			−.21	−.29			−.23	−.29		
	Common	.03	.18	−.26		.03	.14	−.24		.04	.15	−.18	
	N		3,439				2,731				2,613		
Books	Consecrated												
	Mass	.18				.16				.16			
	Specialized	−.18	−.18			−.19	−.20			−.18	−.15		
	Common	.16	.13	−.30		.18	.15	−.26		.16	.13	−.27	
	N		1,599				1,338				1,233		

domain. While replicating all models three times is beyond the scope of this appendix, we focus on model 2 as an illustrative example (given that model 2 includes all taste effects and examines all time periods simultaneously). What this means in practice is that in one replication model the exogenous and endogenous measures focus solely on movie tastes; for the next replication model, they focus solely on music tastes; and so on. Results from these analyses are presented in table A2.

TABLE A2
Taste Effects from Model 2 and Three Replication Models Considering Each Domain Separately

	Model 2	Model A1	Model A2	Model A3
Individual effects:				
Consecrated culture	−.61	−1.37	−1.22	2.96*
	(.43)	(1.05)	(.63)	(1.21)
Mass culture	−.81*	−1.04	−1.37**	−.92
	(.34)	(.72)	(.46)	(1.25)
Specialized culture	−.01	−.82	−.67	1.93
	(.37)	(.83)	(.54)	(1.06)
Common culture	2.15***	2.87**	2.37**	3.10*
	(.52)	(.96)	(.83)	(1.37)
Movies	−1.06***	−.34		
	(.31)	(.50)		
Music	.17		.52	
	(.25)		(.34)	
Books	.73			−1.11
	(.41)			(.70)
Dyadic effects:				
Shared consecrated culture	3.35	10.49*	4.78	−7.82
	(2.41)	(4.64)	(3.98)	(4.51)
Shared mass culture	−3.41	−6.38	−1.92	−6.28
	(1.90)	(3.94)	(2.55)	(4.35)
Shared specialized culture	14.07	40.93**	6.44	−14.41
	(8.22)	(13.64)	(11.49)	(22.63)
Shared common culture	−3.04	−2.35	−5.13	.90
	(2.11)	(3.97)	(3.00)	(5.22)
Shared movies	1.85	−.05		
	(2.45)	(3.40)		
Shared music	2.81		3.41	
	(1.87)		(2.28)	
Shared books	.81			1.95
	(2.89)			(4.60)
Taste similarity (%)	1.94*	2.46***	1.77**	2.44***
	(.77)	(.48)	(.55)	(.47)
Domain	All	Movies	Music	Books

Note.—$N = 520$ for all models. Exogenous and endogenous codes in each model refer only to the domain that is the focus of that model. The "taste similarity" measure still incorporates all domains. All results are multiplied by 100. Data are presented as coefficient (SE).

* $P < .05$.

** $P < .01$.

*** $P < .001$.

Findings suggest there is some important variation among domains that is masked by our aggregate measures. In particular, tastes in consecrated culture books (model A3) have generalized conversion value, but tastes in consecrated culture movies and music do not. Further, dyadic conversion for consecrated and specialized culture tastes is only statistically significant for movies (model A1). Other results are more consistent. Specifically, we find the generalized conversion effect of common culture tastes is statistically significant across all domains, and while the generalized inhibition effect of mass culture tastes is only significant for music (model A2), the size and direction of this effect is similar across all models.

One challenge of this exercise is that, in cases where results vary, it is difficult to assess, for example, whether our coding for consecrated culture movies and consecrated culture books captures slightly different constructs or whether consecrated culture tastes in movies and in books actually convey distinct social signals. As with any composite measure, this is one advantage of aggregating domains: the impact on results of potential variation across individual measures is attenuated. However, we should be mindful of this preliminary evidence that interactions between meanings and domains could be a potentially valuable path for future exploration.

Endogenous Thresholds

As described in the main text, we operationalized the endogenous meaning of each taste by selecting two thresholds of local popularity: a taste was considered "specialized" if it was expressed by fewer than .5% of students at a given wave and "common" if it was expressed by more than 5% of students. How robust are results to alternative thresholds? To answer this question, we again replicated model 2, each time substituting a different threshold for one of our endogenous measures: in model A4 we replaced the threshold for specialized tastes with .25% and in model A5 with 1% (respectively, half and double the original value), and in model A6 we replaced the threshold for common tastes with 2.5% and in model A7 with 10% (respectively, half and double the original value). Results are presented in table A3.

Findings from these replications are reassuring. Specifically, in all cases where there was a significant taste effect in model 2, this effect continued to be statistically significant in all four of the replication models. Additionally, we find that in some cases new effects appeared that were not present in our original model. For instance, when the threshold for common culture tastes is reduced to 2.5%, we find a significant dyadic aversion effect for shared common culture tastes and a significant dyadic conversion effect for shared tastes in music, and when the threshold for common culture tastes is raised to 10%, we find significant generalized conversion effects for tastes in music and in books. Collectively, these findings indicate that different thresholds for endogenous measures may illuminate some variation in conversion dynamics. However, the central conclusions from model 2 are robust across multiple operationalizations.

TABLE A3
Taste Effects from Model 2 and Four Replication Models with Varying Thresholds for Endogenous Measures

	Model 2	Model A4	Model A5	Model A6	Model A7
Individual effects:					
Consecrated culture	−.61	−.62	−.63	−.64	−.53
	(.43)	(.44)	(.44)	(.43)	(.43)
Mass culture	−.81*	−.84*	−.83*	−.75*	−.84*
	(.34)	(.34)	(.34)	(.34)	(.35)
Specialized culture	−.01	−.15	−.09	−.10	−.50
	(.37)	(.42)	(.39)	(.45)	(.32)
Common culture	2.15***	2.04***	2.07***	1.23*	3.38***
	(.52)	(.48)	(.56)	(.49)	(.95)
Movies	−1.06***	−1.02***	−1.01**	−1.07**	−.70*
	(.31)	(.28)	(.36)	(.36)	(.29)
Music	.17	.22	.23	.14	.55**
	(.25)	(.21)	(.31)	(.33)	(.21)
Books	.73	.80*	.79	.81	1.07**
	(.41)	(.40)	(.46)	(.44)	(.39)
Dyadic effects:					
Shared consecrated culture	3.35	3.37	3.41	3.28	3.39
	(2.41)	(2.41)	(2.36)	(2.38)	(2.36)
Shared mass culture	−3.41	−3.44	−3.37	−3.19	−3.15
	(1.90)	(1.85)	(1.85)	(1.87)	(1.85)
Shared specialized culture	14.07	18.12	7.75	9.83	14.45
	(8.22)	(17.63)	(4.96)	(8.69)	(8.15)
Shared common culture	−3.04	−3.44	−2.55	−5.74*	−4.52
	(2.11)	(2.05)	(2.18)	(2.71)	(2.32)
Shared movies	1.85	2.30	1.39	5.08	1.26
	(2.45)	(2.42)	(2.58)	(3.08)	(2.20)
Shared music	2.81	3.23	2.35	5.96*	2.14
	(1.87)	(1.78)	(1.94)	(2.57)	(1.58)
Shared books	.81	1.27	.23	3.84	.20
	(2.89)	(2.84)	(3.04)	(3.39)	(2.67)
Taste similarity (%)	1.94*	1.93*	1.94**	2.02**	1.97**
	(.77)	(.76)	(.75)	(.76)	(.76)
Threshold (%):					
Specialized (less than)	.50	.25	1.00	.50	.50
Common (greater than)	5.00	5.00	5.00	2.50	10.00

Note.—$N = 520$ for all models. All results are multiplied by 100. Data are presented as coefficient (SE).
* $P < .05$.
** $P < .01$.
*** $P < .001$.

APPENDIX B

Modeling Social Networks

Rivera et al. (2010) have documented the rise of network research in top sociology journals over the preceding 40 years. By now, most sociologists are therefore familiar with the fundamental prioritization of relationships that characterizes network research. Beyond this grasp of what network analysts study, however, much less attention is generally paid to how they go about their research in ways that are occasionally very different from mainstream conventions. In this appendix, we wish to clarify and make explicit two such differences that are relevant to this article. These have to do with model building and goodness of fit in the context of stochastic actor-based models.

Model Building

In their introduction to stochastic actor-based models, Snijders et al. (2010) give special attention to model building: "For actor-based models for network dynamics, information-theoretic model selection criteria have not yet generally been developed. . . . Currently the best possibility is to use ad hoc stepwise procedures, combining forward steps (where effects are added to the model) with backward steps (where effects are deleted). The steps can be based on significance tests for the various effects that may be included in the model" (p. 50). Snijders et al. then proceed to outline a series of guideline for practitioners to follow. The approach we describe above, by which we reached the "final" models in this article, attempts to synthesize as many as possible of these guidelines into a straightforward, transparent, and replicable procedure.

This approach may appear strikingly different from mainstream conventions in quantitative social science—where inductive exploration is often discouraged and significance tests should technically be altered to accommodate repeated model runs and deletion of nonsignificant effects. To our understanding, the underlying motivation behind this difference is network analysts' emphasis on fit—where even an effect that is not statistically significant may still be "important" in the sense of achieving a good-fitting model. To date, however, there seems to remain a gap between methodologists' articulation of these principles (often in specialty journals) and their implementation in mainstream research. A large variety of applications of stochastic actor-based models are compiled on the Siena webpage (www.stats.ox.ac.uk/~snijders/siena). For model-building illustrations that are similar to ours in spirit but based instead on exponential random graph models (a technique for cross-sectional analysis), we recommend Goodreau's (2007) examination of a large network from the National Longitudinal Study of Adolescent

Health and also Wimmer and Lewis's (2010) analysis of wave 1 of the same data set used here.

Goodness of Fit

If recommended approaches to network model building often prioritize fit, what criteria of evaluation should be used? Following again the advice in the methodological literature (and also the RSiena manual; Ripley et al. 2016), our approach in the main text utilizes t-tests of individual coefficients and also the score-type goodness-of-fit tests described by Schweinberger (2012). Other, more holistic approaches are also available. An increasingly common technique is to compare the observed network to simulated networks from the final model with respect to macrolevel network properties that are not explicitly modeled (cf. Hunter, Goodreau, and Handcock 2008). One particularly important property is the "degree distribution"—in our case, the distribution of the quantity of friendships over students.

In figure B1, we compare the cumulative degree distribution of the actually observed network (in 10-degree intervals) with simulations based on each of our four "final" models (models 4, 6, 8, and 10). Preliminary visual inspection suggests the simulated and observed distributions are rather similar, although quantitative summaries are less optimistic (all P-values based on Monte Carlo Mahalanobis distance tests are less than .05, which in this case suggests worse fit) and the centered, more fine-grained plots in the right-hand column reveal substantial deviation between simulations and observations. Because our network is considerably larger than many applications of stochastic actor-based models, fit may be generally poorer; larger networks may contain greater internal heterogeneity that current models are unable to satisfactorily represent (we thank Tom Snijders for helpful feedback on this issue). We provide these plots for the sake of transparency and in hope that they will stimulate further development.

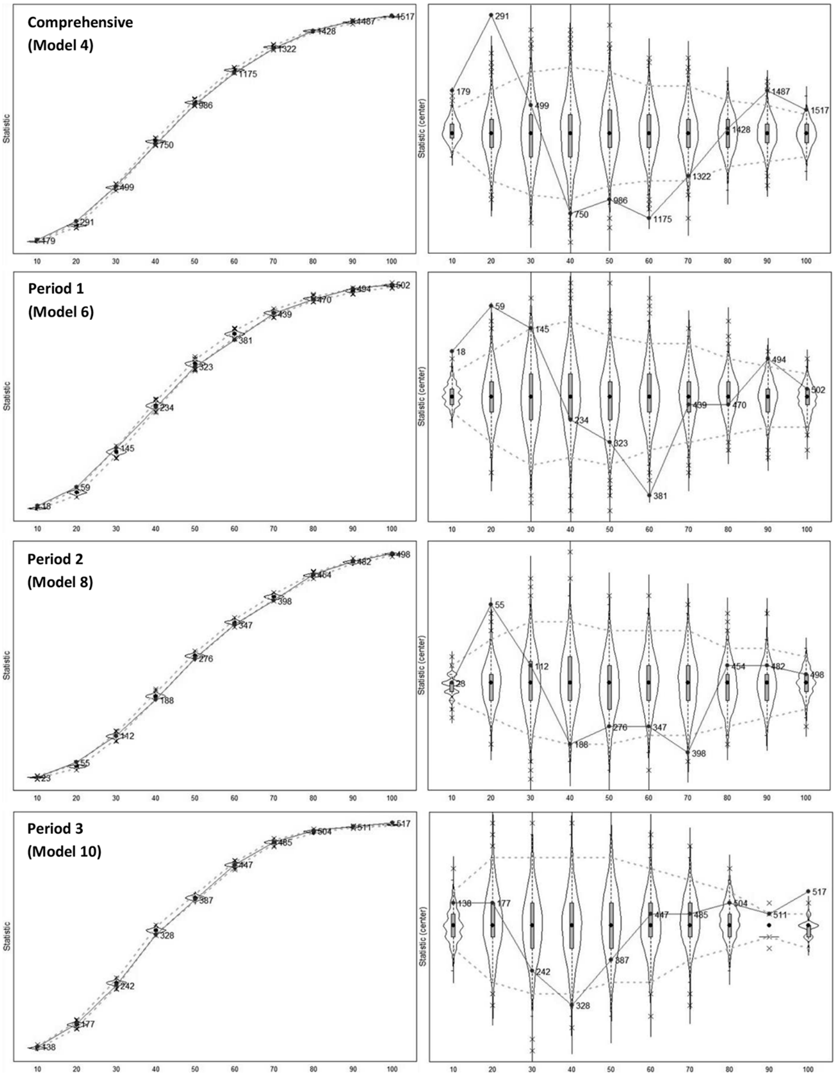


FIG. B1.—Goodness-of-fit plots for models 4, 6, 8, and 10. The solid line represents the cumulative degree distribution of the observed network, aggregated into 10-degree intervals (and presented here for a maximum degree of 100, which represents over 95% of students in all models). Violin plots represent network simulations in phase 3 of the estimation algorithm. Plots in the right column are centered to enhance visibility. Statistics for the comprehensive model (model 4) are aggregated across all periods.

REFERENCES

Adamic, Lada A., and Eytan Adar. 2003. "Friends and Neighbors on the Web." *Social Networks* 25:211–230.

adams, jimi, Katherine Faust, and Gina S. Lovasi. 2012. "Capturing Context: Integrating Spatial and Social Network Analyses." *Social Networks* 34:1–5.

Alderson, Arthur S., Azamat Junisbai, and Isaac Heacock. 2007. "Social Status and Cultural Consumption in the United States." *Poetics* 35:191–212.

Alexander, Jeffrey and Philip Smith. 2001. "The Strong Program in Cultural Theory: Elements of a Structural Hermeneutics." Pp. 135–50 in *Handbook of Sociological Theory*, edited by J. H. Turner. New York: Kluwer Academic.

Allen, Michael Patrick, and Anne E. Lincoln. 2004. "Critical Discourse and the Cultural Consecration of American Films." *Social Forces* 82:871–94.

Aral, Sinan, and Dylan Walker. 2012. "Identifying Influential and Susceptible Members of Social Networks." *Science* 337:337–41.

Arnaboldi, Valerio, Andrea Guazzini, and Andrea Passarella. 2013. "Egocentric Online Social Networks: Analysis of Key Features and Prediction of Tie Strength in Facebook." *Computer Communications* 36:1130–44.

Axelrod, Robert. 1997. "The Dissemination of Culture: A Model with Local Convergence and Global Polarization." *Journal of Conflict Resolution* 41:203–26.

Baumann, Shyon. 2001. "Intellectualization and Art World Development: Film in the United States." *American Sociological Review* 66:404–26.

Bearman, Peter. 1997. "Generalized Exchange." *American Journal of Sociology* 102: 1383–1415.

Benediktsson, Mike Owen. 2012. "Bridging and Bonding in the Academic Melting Pot: Cultural Resources and Network Diversity." *Sociological Forum* 27:46–69.

Bennett, Tony, Mike Savage, Elizabeth Silva, Alan Warde, Modesto Gayo-Cal, and David Wright. 2009. *Culture, Class, Distinction*. London: Routledge.

Block, Per, and Thomas Grund. 2014. "Multidimensional Homophily in Friendship Networks." *Network Science* 2:189–212.

Borgatti, Stephen P., Ajay Mehra, Daniel J. Brass, and Giuseppe Labianca. 2009. "Network Analysis in the Social Sciences." *Science* 323:892–95.

Bourdieu, Pierre. 1984. *Distinction: A Social Critique of the Judgement of Taste*. Translated by R. Nice. Cambridge, Mass.: Harvard University Press.

———. 1986. "The Forms of Capital." Pp. 241–58 in *Handbook of Theory and Research for the Sociology of Education*, edited by J. G. Richardson. New York: Greenwood Press.

Breiger, Ronald L. 1974. "The Duality of Persons and Groups." *Social Forces* 53:181–90.

Bryson, Bethany. 1996. "'Anything but Heavy Metal': Symbolic Exclusion and Musical Dislikes." *American Sociological Review* 61:884–99.

Bukodi, Erzsébet. 2007. "Social Stratification and Cultural Consumption in Hungary: Book Readership." *Poetics* 35:112–31.

Campbell, Karen E., Peter V. Marsden, and Jeanne S. Hurlbert. 1986. "Social Resources and Socioeconomic Status." *Social Networks* 8:97–117.

Carley, Kathleen. 1991. "A Theory of Group Stability." *American Sociological Review* 56:331–54.

Cartwright, Dorwin, and Frank Harary. 1956. "Structural Balance: A Generalization of Heider's Theory." *Psychological Review* 63:277–93.

Chan, Tak Wing, and John H. Goldthorpe. 2005. "The Social Stratification of Theatre, Dance and Cinema Attendance." *Cultural Trends* 14:193–212.

Coser, Lewis A. 1974. *Greedy Institutions: Patterns of Undivided Commitment*. New York: Free Press.

Dahlander, Linus, and Daniel A. McFarland. 2013. "Ties That Last: Tie Formation and Persistence in Research Collaborations over Time." *Administrative Science Quarterly* 58:69–110.

de Klepper, Maurits, Ed Sleebos, Gerhard van de Bunt, and Filip Agneessens. 2010. "Similarity in Friendship Networks: Selection or Influence? The Effect of Constraining Contexts and Non-Visible Individual Attributes." *Social Networks* 32:82–90.

DiMaggio, Paul. 1982. "Cultural Capital and School Success: The Impact of Status Culture Participation on the Grades of U.S. High School Students." *American Sociological Review* 47:189–201.

———. 1987. "Classification in Art." *American Sociological Review* 52:440–55.

———. 2004. "Gender, Networks, and Cultural Capital." *Poetics* 32:99–103.

DiMaggio, Paul, and Filiz Garip. 2012. "Network Effects and Social Inequality." *Annual Review of Sociology* 38:93–118.

DiMaggio, Paul, and John Mohr. 1985. "Cultural Capital, Educational Attainment, and Marital Selection." *American Journal of Sociology* 90:1231–61.

Edelmann, Achim, and Stephen Vaisey. 2014. "Cultural Resources and Cultural Distinction in Networks." *Poetics* 46:22–37.

Eliasoph, Nina, and Paul Lichterman. 2003. "Culture in Interaction." *American Journal of Sociology* 108:735–94.

Ellison, Nicole B., Charles Steinfield, and Cliff Lampe. 2007. "The Benefits of Facebook 'Friends': Social Capital and College Students' Use of Online Social Network Sites." *Journal of Computer-Mediated Communication* 12:1143–68.

Emirbayer, Mustafa, and Jeff Goodwin. 1994. "Network Analysis, Culture, and the Problem of Agency." *American Journal of Sociology* 99:1411–54.

Erickson, Bonnie H. 1996. "Culture, Class, and Connections." *American Journal of Sociology* 102:217–51.

Feld, Scott L. 1981. "The Focused Organization of Social Ties." *American Journal of Sociology* 86:1015–35.

———. 1982. "Social Structural Determinants of Similarity among Associates." *American Sociological Review* 47:797–801.

Fine, Gary Alan, and Corey D. Fields. 2008. "Culture and Microsociology: The Anthill and the Veldt." *Annals of the American Academy of Political and Social Science* 619:130–48.

Fischer, Claude S. 1982. "What Do We Mean by 'Friend'? An Inductive Study." *Social Networks* 3:287–306.

Golder, Scott A., and Michael W. Macy. 2014. "Digital Footprints: Opportunities and Challenges for Online Social Research." *Annual Review of Sociology* 40:129–52.

Gonçalves, Bruno, Nicola Perra, and Alessandro Vespignani. 2011. "Modeling Users' Activity on Twitter Networks: Validation of Dunbar's Number." *PLoS ONE* 6:e22656.

Goodreau, Steven M. 2007. "Advances in Exponential Random Graph (p*) Models Applied to a Large Social Network." *Social Networks* 29:231–48.

Goodreau, Steven M., James A. Kitts, and Martina Morris. 2009. "Birds of a Feather, or Friend of a Friend? Using Exponential Random Graph Models to Investigate Adolescent Social Networks." *Demography* 46:103–25.

Griswold, Wendy, Terry McDonnell, and Nathan Wright. 2005. "Reading and the Reading Class in the Twenty-First Century." *Annual Review of Sociology* 31:127–41.

Hagan, John. 1991. "Destiny and Drift: Subcultural Preferences, Status Attainments, and the Risks and Rewards of Youth." *American Sociological Review* 56:567–82.

Harrington, Brooke, and Gary Alan Fine. 2006. "Where the Action Is: Small Groups and Recent Developments in Sociological Theory." *Small Group Research* 37:4–19.

Hughes, Michael, and Richard A. Peterson. 1983. "Isolating Cultural Choice Patterns in the U.S. Population." *American Behavioral Scientist* 26:459–78.

Huisman, Mark, and Christian Steglich. 2008. "Treatment of Non-Response in Longitudinal Network Studies." *Social Networks* 30:297–308.

Hunter, David R., Steven M. Goodreau, and Mark S. Handcock. 2008. "Goodness of Fit of Social Network Models." *Journal of the American Statistical Association* 103:248–58.

Jacobs, Mark D., and Lyn Spillman. 2005. "Cultural Sociology at the Crossroads of the Discipline." *Poetics* 33:1–14.

Kalmijn, Matthijs. 1994. "Assortative Mating by Cultural and Economic Occupational Status." *American Journal of Sociology* 100:422–52.

Katz-Gerro, Tally. 1999. "Cultural Consumption and Social Stratification: Leisure Activities, Musical Tastes, and Social Location." *Sociological Perspectives* 42:627–46.

Kaufman, Jason. 2004. "Endogenous Explanation in the Sociology of Culture." *Annual Review of Sociology* 30:335–57.

Kaufman, Jason, and Jay Gabler. 2004. "Cultural Capital and the Extracurricular Activities of Girls and Boys in the College Attainment Process." *Poetics* 32:145–68.

Kossinets, Gueorgi, and Duncan J. Watts. 2009. "Origins of Homophily in an Evolving Social Network." *American Journal of Sociology* 115:405–50.

Lahire, Bernard. 2008. "The Individual and the Mixing of Genres: Cultural Dissonance and Self-Distinction." *Poetics* 36:166–88.

Lamont, Michèle, and Annette Lareau. 1988. "Cultural Capital: Allusions, Gaps and Glissandos in Recent Theoretical Developments." *Sociological Theory* 6:153–68.

Lewis, Kevin. 2015*a*. "How Networks Form: Homophily, Opportunity, and Balance." In *Emerging Trends in the Social and Behavioral Sciences*, edited by R. Scott and S. Kosslyn. Hoboken, N.J.: John Wiley & Sons.

———. 2015*b*. "Three Fallacies of Digital Footprints." *Big Data and Society* 2. https://doi.org/10.1177/2053951715602496.

Lewis, Kevin, Marco Gonzalez, and Jason Kaufman. 2012. "Social Selection and Peer Influence in an Online Social Network." *Proceedings of the National Academy of Sciences of the United States of America* 109:68–72.

Lewis, Kevin, Jason Kaufman, and Nicholas Christakis. 2008*a*. "The Taste for Privacy: An Analysis of College Student Privacy Settings in an Online Social Network." *Journal of Computer-Mediated Communication* 14:79–100.

Lewis, Kevin, Jason Kaufman, Marco Gonzalez, Andreas Wimmer, and Nicholas Christakis. 2008*b*. "Tastes, Ties, and Time: A New Social Network Dataset Using Facebook.com." *Social Networks* 30:330–42.

Lieberson, Stanley. 2000. *A Matter of Taste: How Names, Fashions, and Culture Change*. New Haven, Conn.: Yale University Press.

Lizardo, Omar. 2006. "How Cultural Tastes Shape Personal Networks." *American Sociological Review* 71:778–807.

Lomi, Alessandro, and Christoph Stadtfeld. 2014. "Social Networks and Social Settings: Developing a Coevolutionary View." *Kölner Zeitschrift für Soziologie und Sozialpsychologie* 66:395–415.

Long, Elizabeth. 2003. *Book Clubs: Women and the Uses of Reading in Everyday Life*. Chicago: University of Chicago Press.

Lopes, Paul. 2006. "Culture and Stigma: Popular Culture and the Case of Comic Books." *Sociological Forum* 21:387–414.

Macdonald, Dwight. 1983. *Against the American Grain*. New York: Da Capo Press.

Mark, Noah. 1998. "Birds of a Feather Sing Together." *Social Forces* 77:453–85.

———. 2003. "Culture and Competition: Homophily and Distancing Explanations for Cultural Niches." *American Sociological Review* 68:319–45.

Marmaros, David, and Bruce Sacerdote. 2006. "How Do Friendships Form?" *Quarterly Journal of Economics* 121:79–119.

Marsden, Peter V. 1987. "Core Discussion Networks of Americans." *American Sociological Review* 52:122–31.

Marsden, Peter V., and Joseph F. Swingle. 1994. "Conceptualizing and Measuring Culture in Surveys: Values, Strategies, and Symbols." *Poetics* 22:269–89.

Mayer, Adalbert, and Steven L. Puller. 2008. "The Old Boy (and Girl) Network: Social Network Formation on University Campuses." *Journal of Public Economics* 92:329–47.

McPherson, J. Miller. 1976. "Theory Trimming." *Social Science Research* 5:95–105.

McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. "Birds of a Feather: Homophily in Social Networks." *Annual Review of Sociology* 27:415–44.

Mercken, L., Tom A. B. Snijders, Christian Steglich, E. Vartiainen, and H. de Vries. 2010. "Dynamics of Adolescent Friendship Networks and Smoking Behavior." *Social Networks* 32:72–81.

Mohr, John W. 1998. "Measuring Meaning Structures." *Annual Review of Sociology* 24:345–70.

Mohr, John W., and Amin Ghaziani. 2014. "Problems and Prospects of Measurement in the Study of Culture." *Theory and Society* 43:225–46.

Ollivier, Michèle. 2008. "Modes of Openness to Cultural Diversity: Humanist, Populist, Practical, and Indifferent." *Poetics* 36:120–47.

Ostrower, Francie. 1998. "The Arts as Cultural Capital among Elites: Bourdieu's Theory Reconsidered." *Poetics* 26:43–53.

Pachucki, Mark A., and Ronald L. Breiger. 2010. "Cultural Holes: Beyond Relationality in Social Networks and Culture." *Annual Review of Sociology* 36:205–24.

Peterson, Richard A. 1992. "Understanding Audience Segmentation: From Elite and Mass to Omnivore and Univore." *Poetics* 21:243–58.

———. 2005. "Problems in Comparative Research: The Example of Omnivorousness." *Poetics* 33:257–82.

Peterson, Richard A., and Roger M. Kern. 1996. "Changing Highbrow Taste: From Snob to Omnivore." *American Sociological Review* 61:900–907.

Popielarz, Pamela A., and Zachary P. Neal. 2007. "The Niche as a Theoretical Tool." *Annual Review of Sociology* 33:65–84.

Ripley, Ruth M., Tom A. B. Snijders, Zsófia Boda, András Vörös, and Paulina Preciado. 2016. "Manual for Rsiena (Version May 28, 2016)." Oxford: University of Oxford, Department of Statistics, Nuffield College.

Rivera, Lauren A. 2012. "Hiring as Cultural Matching: The Case of Elite Professional Service Firms." *American Sociological Review* 77:999–1022.

Rivera, Mark T., Sara B. Soderstrom, and Brian Uzzi. 2010. "Dynamics of Dyads in Social Networks: Assortative, Relational, and Proximity Mechanisms." *Annual Review of Sociology* 36:91–115.

Robins, Garry, Pip Pattison, Yuval Kalish, and Dean Lusher. 2007. "An Introduction to Exponential Random Graph (P*) Models for Social Networks." *Social Networks* 29:173–91.

Robinson, John P., and Geoffrey Godbey. 1997. *Time for Life: The Surprising Ways Americans Use Their Time*. University Park: Pennsylvania State University Press.

Roy, William G., and Timothy J. Dowd. 2010. "What Is Sociological about Music?" *Annual Review of Sociology* 36:183–203.

Schultz, Jennifer, and Ronald L. Breiger. 2010. "The Strength of Weak Culture." *Poetics* 38:610–24.

Schweinberger, Michael. 2012. "Statistical Modeling of Network Panel Data: Goodness-of-Fit." *British Journal of Statistical and Mathematical Psychology* 65:263–81.

Selfhout, Maarten H. W., Susan J. T. Branje, Tom F. M. ter Bogt, and Wim H. J. Meeus. 2009. "The Role of Music Preferences in Early Adolescents' Friendship Formation and Stability." *Journal of Adolescence* 32:95–107.

Snijders, Tom A. B. 1996. "Stochastic Actor-Oriented Dynamic Network Analysis." *Journal of Mathematical Sociology* 21:149–72.

———. 2001. "The Statistical Evaluation of Social Network Dynamics." *Sociological Methodology* 31:361–95.

———. 2005. "Models for Longitudinal Network Data." Pp. 215–47 in *Models and Methods in Social Network Analysis*, edited by P. J. Carrington, J. Scott, and S. Wasserman. Cambridge: Cambridge University Press.

———. 2011. "Statistical Models for Social Networks." *Annual Review of Sociology* 37:131–53.

Snijders, Tom A. B., and Patrick Doreian. 2010. “Introduction to the Special Issue on Network Dynamics.” *Social Networks* 32:1–3.
———. 2012. “Introduction to the Special Issue on Network Dynamics (Part 2).” *Social Networks* 34:289–90.
Snijders, Tom A. B., Gerhard van de Bunt, and Christian Steglich. 2010. “Introduction to Stochastic Actor-Based Models for Network Dynamics.” *Social Networks* 32:44–60.
Steglich, Christian, Tom A. B. Snijders, and Michael Pearson. 2010. “Dynamic Networks and Behavior: Separating Selection from Influence.” *Sociological Methodology* 40:329–93.
Steglich, Christian, Tom A. B. Snijders, and Patrick West. 2006. “Applying Siena: An Illustrative Analysis of the Coevolution of Adolescents’ Friendship Networks, Taste in Music, and Alcohol Consumption.” *Methodology* 2:48–56.
Swidler, Ann. 1986. “Culture in Action: Symbols and Strategies.” *American Sociological Review* 51:273–86.
Vaisey, Stephen. 2009. “Motivation and Justification: A Dual-Process Model of Culture in Action.” *American Journal of Sociology* 114:1675–1715.
Vaisey, Stephen, and Omar Lizardo. 2010. “Can Cultural Worldviews Influence Network Composition?” *Social Forces* 88:1595–1618.
van Duijn, Marijtje A. J., Evelien P. H. Zeggelink, Mark Huisman, Frans N. Stokman, and Frans W. Wasseur. 2003. “Evolution of Sociology Freshmen into a Friendship Network.” *Journal of Mathematical Sociology* 27:153–91.
van Venrooij, Alex. 2009. “The Aesthetic Discourse Space of Popular Music: 1985–86 and 2004–05.” *Poetics* 37:315–32.
Verbrugge, Lois M. 1977. “The Structure of Adult Friendship Choices.” *Social Forces* 56:576–97.
Warde, Alan, and Modesto Gayo-Cal. 2009. “The Anatomy of Cultural Omnivorousness: The Case of the United Kingdom.” *Poetics* 37:119–45.
Werner, Carol, and Pat Parmelee. 1979. “Similarity of Activity Preferences among Friends: Those Who Play Together Stay Together.” *Social Psychology Quarterly* 42:62–66.
Wimmer, Andreas, and Kevin Lewis. 2010. “Beyond and Below Racial Homophily: ERG Models of a Friendship Network Documented on Facebook.” *American Journal of Sociology* 116:583–642.
Zavisca, Jane. 2005. “The Status of Cultural Omnivorism: A Case Study of Reading in Russia.” *Social Forces* 84:1233–55.
Zywica, Jolene, and James Danowski. 2008. “The Faces of Facebookers: Investigating Social Enhancement and Social Compensation Hypotheses; Predicting Facebook and Offline Popularity from Sociability and Self-Esteem, and Mapping the Meanings of Popularity with Semantic Networks.” *Journal of Computer-Mediated Communication* 14:1–34.